Definite Integration

21. Let (a, b) be the point of intersection of the curve $x^2 = 2y$ and the straight line $y - 2x - 6 = 0$ in the second quadrant.

Then the integral $I = \int_a^b \frac{9x^2}{1+5^x}dx$ is equal to :

माना वक्र $x^2 = 2y$ और सरल रेखा $y - 2x - 6 = 0$ का दूसरे चतुर्थांश में प्रतिच्छेदन बिन्दु (a, b) है। तो समाकलन $I = \int_a^b \frac{9x^2}{1+5^x}dx$ बराबर है :

(A) 24 (B) 18 (C) 21 (D) 27

Ans. A

Sol. $x^2 = 2y$ and $y - 2x - 6 = 0$

$$\frac{x^2}{2} - 2x - 6 = 0$$

$x^2 - 4x - 12 = 0$

$x^2 - 6x + 2x - 12 = 0$

$x(x - 6) + 2(x - 6) = 0$

$(x - 6)(x + 2) = 0$

Point of intersection are (6, 18) and (–2, 2)

(–2, 2) is in second quadrant

$a = -2, b = 2$

$$I = \int_{-2}^{2} \frac{9x^2}{1+5^x}dx \quad ...(1)$$

$$I = \int_{-2}^{2} \frac{9x^2}{1+5^{-x}}dx \quad ...(2)$$

Adding (1) and (2)

$$2I = \int_{-2}^{2} 9x^2dx$$

$$I = 9\int_{0}^{2} x^2dx$$

$$I = 9\left(\frac{x^3}{3}\right)_0^2 \Rightarrow I = 24$$

22. $4\int_0^1\left(\frac{1}{\sqrt{3+x^2}+\sqrt{1+x^2}}\right)dx-3\log_e\left(\sqrt{3}\right)$ is equal to :

$4\int_0^1\left(\frac{1}{\sqrt{3+x^2}+\sqrt{1+x^2}}\right)dx-3\log_e\left(\sqrt{3}\right)$ बराबर है :

(A) $2+\sqrt{2}-\log_e\left(1+\sqrt{2}\right)$ (B) $2+\sqrt{2}+\log_e\left(1+\sqrt{2}\right)$

(C) $2-\sqrt{2}+\log_e\left(1+\sqrt{2}\right)$ (D) $2-\sqrt{2}-\log_e\left(1+\sqrt{2}\right)$

Ans. D

Sol. $I=4\int_0^1\frac{1}{\sqrt{3+x^2}+\sqrt{1+x^2}}dx$

$=2\int_0^1\sqrt{3+x^2}-\sqrt{1+x^2}dx$

$=2\left[\int_0^1\sqrt{3+x^2}dx-\int_0^1\sqrt{1+x^2}dx\right]$

$=2\left[\left(\frac{1}{2}x\sqrt{x^2+3}+\frac{3}{2}\ell n\left|\sqrt{3+x^2}+x\right|\right)-\left(\frac{1}{2}x\sqrt{1+x^2}+\frac{1}{2}\ell n\left|\sqrt{1+x^2}+x\right|\right)\right]_0^1$

$=2\left[\left(1+\frac{3}{2}\ell n3-\frac{3}{2}\ell n\sqrt{3}\right)-\left(\frac{\sqrt{2}}{2}+\frac{1}{2}\ell n\left(\sqrt{2}+1\right)\right)\right]$

$=2\left(1+\frac{3}{4}\ell n3-\frac{1}{\sqrt{2}}-\frac{1}{2}\ell n\left(\sqrt{2}+1\right)\right)$

$=3\ell n\sqrt{3}+2-\sqrt{2}-\ell n\left(\sqrt{2}+1\right)$

$I-3\ell n\sqrt{3}=2-\sqrt{2}-\ell n\left(\sqrt{2}+1\right)$

23. Let the domain of the function $f(x)=\log_2\log_4\log_6(3+4x-x^2)$ be (a, b). If $\int_0^{b-a}\left[x^2\right]dx=p-\sqrt{q}-\sqrt{r}$, p, q, r ∈ N, gcd(p, q, r) = 1, where [·] is the greatest integer function, then p + q + r is equal to :

माना फलन $f(x)=\log_2\log_4\log_6(3+4x-x^2)$ का प्रांत (a, b) है। यदि $\int_0^{b-a}\left[x^2\right]dx=p-\sqrt{q}-\sqrt{r}$, p, q, r ∈ N, gcd(p, q, r) = 1 हैं, जहाँ [·] अधिकतम पूर्णांक फलन है, तो p + q + r बराबर है :

(A) 11 (B) 10 (C) 8 (D) 9

Ans. B

Sol. $\log_4 \log_6(3 + 4x - x^2) > 0$

$\log_6(3 + 4x - x^2) > 1$

$3 + 4x - x^2 > 6$

$x \in (1, 3)$

$a = 1, b = 3$

$$I = \int_0^2 \left[x^2\right] dx = \int_0^1 \left[x^2\right] dx + \int_1^{\sqrt{2}} \left[x^2\right] dx + \int_{\sqrt{2}}^{\sqrt{3}} \left[x^2\right] dx + \int_{\sqrt{3}}^2 \left[x^2\right] dx$$

$$= 5 - \sqrt{2} - \sqrt{3}$$

$p + q + r = 10$

24. The integral $\int_0^{\pi} \frac{8x dx}{4\cos^2 x + \sin^2 x}$ is equal to :

समाकलन $\int_0^{\pi} \frac{8x dx}{4\cos^2 x + \sin^2 x}$ बराबर है :

(A) $\frac{3\pi^2}{2}$ (B) π^2 (C) $4\pi^2$ (D) $2\pi^2$

Ans. D

Sol. $\because I = \int_0^{\pi} \frac{8x}{4\cos^2 x + \sin^2 x} dx$ ...(1)

$$I = \int_0^{\pi} \frac{8(\pi - x)}{4\cos^2 x + \sin^2 x} dx$$ ...(2)

(A) + (2)

$$2I = \int_0^{\pi} \frac{8\pi}{4\cos^2 x + \sin^2 x} dx$$

$$I = 4\pi \int_0^{\pi} \frac{1}{1 + 3\cos^2 x} dx$$

$$I = 8\pi\int_0^{\frac{\pi}{2}} \frac{1}{1+3\cos^2 x}dx$$

$$I = 8\pi\int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{\sec^2 x+3}dx$$

$$I = 8\pi\int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{4+\tan^2 x}dx$$ Put $\tan x = t$

$\sec^2 x\, dx = dt$

$$I = \int_0^{\infty} \frac{dt}{t^2+2^2}$$

$$I = 4\pi\left(\tan^{-1}\left(\frac{t}{2}\right)\right)_0^{\infty} \Rightarrow U = 4\pi\times\frac{\pi}{2} = 2\pi^2$$

25. The value of $\int_{-1}^{1} \frac{(1+\sqrt{|x|-x})e^x+(\sqrt{|x|-x})e^{-x}}{e^x+e^{-x}}dx$ is equal to :

$\int_{-1}^{1} \frac{(1+\sqrt{|x|-x})e^x+(\sqrt{|x|-x})e^{-x}}{e^x+e^{-x}}dx$ का मान बराबर होगा।

(A) $2+\frac{2\sqrt{2}}{3}$ (B) $3-\frac{2\sqrt{2}}{3}$ (C) $1-\frac{2\sqrt{2}}{3}$ (D) $1+\frac{2\sqrt{2}}{3}$

Ans. D

Sol. $$I = \int_{-1}^{1} \frac{\left(1+\sqrt{|x|-x}\right)e^x+\left(\sqrt{|x|-x}\right)e^{-x}}{e^x+e^{-x}}dx \quad ...(1)$$

Apply King's Property

$$I = \int_{-1}^{1} \frac{\left(1+\sqrt{|x|+x}\right)\cdot e^{-x}+\left(\sqrt{|x|+x}\right)e^x dx}{e^{-x}+e^x} \quad ...(2)$$

Add (1) & (2)

$$2I = \int_{-1}^{1} \frac{e^x\left\{1+\sqrt{|x|-x}+\sqrt{|x|+x}\right\}+e^x\left\{\sqrt{|x|-x}+1+\sqrt{|x|+x}\right\}dx}{e^x+e^{-x}}$$

$$= \int_{-1}^{1} \frac{\left(1+\sqrt{|x|-x}+\sqrt{|x|+x}\right)\left(e^{x}+e^{-x}\right)dx}{e^{x}+e^{-x}}$$

$$2I = \int_{-1}^{1} \left(1+\sqrt{|x|-x}+\sqrt{|x|+x}\right)dx$$

Even function

$$I = \int_{0}^{1} \left(1+\sqrt{|x|-x}+\sqrt{|x|+x}\right)dx$$

$$= \int_{0}^{1} \left(1+\sqrt{2x}\right)dx$$

$$I = 1+\sqrt{2}\cdot\frac{2x^{\frac{3}{2}}}{3}\Bigg|_{0}^{1}$$

$$= 1+\frac{2\sqrt{2}}{3}$$

26. Let $f(x)+2f\left(\frac{1}{x}\right)=x^2+5$ and $2g(x)-3g\left(\frac{1}{2}\right)=x, x>0$. If $\alpha=\int_1^2 f(x)dx$, and $\beta=\int_1^2 g(x)dx$, then the value of $9\alpha+\beta$ is :

माना $f(x)+2f\left(\frac{1}{x}\right)=x^2+5$ और $2g(x)-3g\left(\frac{1}{2}\right)=x, x>0$ हैं। यदि $\alpha=\int_1^2 f(x)dx$, और $\beta=\int_1^2 g(x)dx$ हैं, तो $9\alpha+\beta$ का मान है :

(A) 0 (B) 10 (C) 11 (D) 1

Ans. C

Sol. $f(x)+2f\left(\frac{1}{x}\right)=x^2+5$

$$2f(x)+f\left(\frac{1}{x}\right)=\frac{1}{x^2}+5\Big\}\times 2$$

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$$-3f(x)=x^2+5-\frac{2}{x^2}-10$$

$$f(x)=\frac{-1}{3}\left(x^2-\frac{2}{x^2}-5\right)$$

$$\alpha=\frac{-1}{3}\left(\frac{x^3}{3}+\frac{2}{x}-5x\right)_1^2$$

$$=-\frac{1}{3}\left(\left(\frac{8}{3}+1-10\right)-\left(\frac{1}{3}+2-5\right)\right)$$

$$=\frac{-1}{3}\left(\frac{7}{3}-6\right)=\frac{-1}{3}\left(\frac{-11}{3}\right)=\frac{11}{9}$$

$$\Rightarrow 2g(x)-3g\left(\frac{1}{2}\right)=x$$

Put $x=\frac{1}{2}\Rightarrow g\left(\frac{1}{2}\right)=\frac{-1}{2}$

$$2g(x)=x-\frac{3}{2}$$

$$g(x)=\frac{1}{2}\left(x-\frac{3}{2}\right)$$

$$\beta=\frac{1}{2}\left(\frac{x^2}{2}-\frac{3}{2}x\right)_1^2$$

$$=\frac{1}{2}\left((2-3)-\left(\frac{1}{2}-\frac{3}{2}\right)\right)$$

$$=\frac{1}{2}\left((-1)-(-1)\right)=0$$

27. The integral $\int_0^{\pi}\frac{(x+3)\sin x}{1+3\cos^2 x}dx$ is equal to :

समाकलन $\int_0^{\pi}\frac{(x+3)\sin x}{1+3\cos^2 x}dx$ बराबर है :

(A) $\frac{\pi}{\sqrt{3}}(\pi+2)$ (B) $\frac{\pi}{\sqrt{3}}(\pi+1)$ (C) $\frac{\pi}{3\sqrt{3}}(\pi+6)$ (D) $\frac{\pi}{2\sqrt{3}}(\pi+4)$

Ans. C

Sol. $I=\int_0^{\pi}\frac{(x+3)\sin x}{1+3\cos^2 x}dx$

$I=\int_0^{\pi}\frac{(\pi-x+3)\sin x}{(1+3\cos^2 x)}dx$

$2I=\int_0^{\pi}\frac{(\pi+6)\sin x.dx}{(1+3\cos^2 x)}=2\int_0^{\frac{\pi}{2}}\frac{(\pi+6)\sin x}{(1+3\cos^2 x)}$

$I=\int_0^{\frac{\pi}{2}}\frac{(\pi+6)\sin x.dx}{(1+3\cos^2 x)}=\frac{\pi}{3\sqrt{3}}(\pi+6)$

$\sqrt{3}\cos x=t$

$\sqrt{3}\sin x=dt$

28. Let f(x) be a positive function and $I_1=\int_{-\frac{1}{2}}^{1}2xf\left(2x(1-2x)\right)dx$ and $I_2=\int_{-1}^{2}f\left(x(1-x)\right)dx$. Then the value of $\frac{I_2}{I_1}$ is equal to :

. माना f(x) एक धनात्मक फलन है तथा $I_1=\int_{-\frac{1}{2}}^{1}2xf\left(2x(1-2x)\right)dx$ और $I_2=\int_{-1}^{2}f\left(x(1-x)\right)dx$ हैं, तो $\frac{I_2}{I_1}$ का मान बराबर है :

(A) 9 (B) 4 (C) 12 (D) 6

Ans. B

Sol. $I_1=\int_{-1/2}^{1}2xf\left(2x(1-2x)\right)dx$

$I_2=\int_{-1}^{2}t\left(ft(1-t)\right)\frac{dt}{2}$ _____(i) (put $2x=t$)

Use prop. V

$I_1=\int_{-1}^{2}(1-t)f(1-t)t)\frac{dt}{2}$ _____(ii)

Equation (1) + (2)

$2I_1=\int_{-1}^{2}f(t(1-t)\frac{dt}{2}$

$$4I_1 = \int_{-1}^{2} f\left(t(1-t)dt = I_2\right.$$

29. The integral $\int_{-1}^{\frac{3}{2}} \left(\left|\pi^2 x \sin(\pi x)\right|\right) dx$ is equal to :

समाकलन $\int_{-1}^{\frac{3}{2}} \left(\left|\pi^2 x \sin(\pi x)\right|\right) dx$ बराबर है :

(A) $1 + 3\pi$ (B) $2 + 3\pi$ (C) $4 + \pi$ (D) $3 + 2\pi$

Ans. A

Sol. $I = \int_{-1}^{3/2} \left|\pi^2 x \sin \pi x\right| dx \cdot = \int_{-1}^{3/2} \pi \mid \pi x \sin \pi x \mid dx$

$I = \int_{-\pi}^{3\pi/2} |t \sin t| \, dt$ (let $\pi x = t \Rightarrow \pi dx = dt$)

$I = 2\int_{0}^{\pi} |t \sin t| \, dt + \int_{\pi}^{3\pi/2} |t \sin t| \, dt$

$I = 2\int_{0}^{\pi} t \sin t dt - \int_{\pi}^{3\pi/2} t \sin t dt$

$I = 2\left(\left(t(-\cos t)_0^{\pi}\right) - \int_0^{\pi} -\cos t dt\right) - \left(\left(t(-\cos t)_{\pi}^{3\pi/2}\right) - \int_{\pi}^{3\pi/2} -\cos t dt\right)$

$I = 2\left(\pi \cdot 1 + (\sin t)_0^{\pi}\right) - \left(+\pi \cdot (-1) + (\sin t)_{\pi}^{3\pi/2}\right)$

$I = 2\pi + \pi + 1 = 1 + 3\pi$

30. Let for $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$, $I_1 = \int_0^{\pi/4} f(x)dx$ and $I_2 = \int_0^{\pi/4} x\, f(x)dx$. then $7I_1 + 12I_2$ is equal to :

माना कि $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$ के लिए, $I_1 = \int_0^{\pi/4} f(x)dx$ और $I_2 = \int_0^{\pi/4} x\, f(x)dx$ है। तब $7I_1 + 12I_2$ बराबर है :

(A) 2 (B) 2π (C) π (D) 1

Ans. D

Sol. $f(x) = (7\tan^6 x - 3\tan^2 x) \cdot \sec^2 x$

$$\therefore I_1 = \int_0^{\frac{\pi}{4}} f(x)\,dx = \int_0^1 \left(7t^6 - 3t^2\right) dt = \left(t^7 - t^3\right)_0^1 = 0$$

Now $$I_2 = \int_0^{\frac{\pi}{4}} xf(x)\,dx = \int_0^1 \frac{\left(7t^6 - 3t^2\right)}{II} \frac{\tan^{-1} t}{I} dt$$

$$= \left(\tan^{-1} t \cdot \left(t^7 - t^3\right)\right)_0^1 - \int_0^1 \left(t^7 - t^3\right) \cdot \frac{1}{1+t^2} dt$$

$$= \int_0^1 \frac{t^3\left(1-t^4\right)}{1+t^2} dt = \int_0^1 t^3\left(1-t^2\right) dt$$

$$= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

Now, $7I_1 + 12I_2 = 1$

31. The value of $\int_{e^2}^{e^4} \frac{1}{x}\left(\frac{e^{\left((\log_e x)^2+1\right)^{-1}}}{e^{\left((\log_e x)^2+1\right)^{-1}} + e^{\left((6-\log_e x)^2+1\right)^{-1}}}\right) dx$ is :

$\int_{e^2}^{e^4} \frac{1}{x}\left(\frac{e^{\left((\log_e x)^2+1\right)^{-1}}}{e^{\left((\log_e x)^2+1\right)^{-1}} + e^{\left((6-\log_e x)^2+1\right)^{-1}}}\right) dx$ का मान है :

(A) 2 (B) $\log_e 2$ (C) e^2 (D) 1

Ans. D

Sol. put $\ell nx = t$

$$I = \int_2^4 \frac{e^{\left(t^2+1\right)^{-1}}}{e^{\left(t^2+1\right)^{-1}} + e^{\left((6-t)^2+1\right)^{-1}}} dt$$

King add

$$2I = \int_2^4 (1)\,dt$$

$I = 1$

32. If $I(m, n) = \int_0^1 x^{m-1}\,(1-x)^{n-1}\,dx$, $m, n > 0$, then $I(9, 14) + I(10, 13)$ is :

यदि $I(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, $m, n > 0$ है, तो $I(9, 14) + I(10, 13)$ है :

(A) $I(9, 13)$ (B) $I(1, 13)$ (C) $I(19, 27)$ (D) $I(9, 1)$

Ans. A

Sol. $I(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$

Let $x = \sin 2\theta$ $dx = 2 \sin\theta \cos\theta \, d\theta$

$$I(m,n) = 2\int_0^{\frac{\pi}{2}} (\sin\theta)^{2m-1} (\cos\theta)^{2n-1} d\theta$$

$$I(9, 14) + I(10, 13) = 2\int_0^{\frac{\pi}{2}} (\sin\theta)^{17} (\cos\theta)^{27} d\theta + 2\int_0^{\frac{\pi}{2}} (\sin\theta)^{19} (\cos\theta)^{25} d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} (\sin\theta)^{17} (\cos\theta)^{25} d\theta$$

$= I(9, 13)$

33. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2 \cos^2 x}{(1+e^x)} dx = \pi(\alpha\pi^2 + \beta)$, $\alpha, \beta \in Z$ then $(\alpha + \beta)^2$ equals :

यदि $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2 \cos^2 x}{(1+e^x)} dx = \pi(\alpha\pi^2 + \beta)$, $\alpha, \beta \in Z$ है तो $(\alpha + \beta)^2$ बराबर है :

(A) 196 (B) 64 (C) 144 (D) 100

Ans. D

Sol. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2 \cos^2 x}{(1+e^x)} dx$ (Apply king property)

$$\int_0^{\frac{\pi}{2}} 96x^2 \cos^2 x = 48\int_0^{\frac{\pi}{2}} x^2 (1+\cos 2x) dx$$

$$48\left[\left(\frac{x^3}{3}\right)_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \underbrace{x^2}_{I}\underbrace{\cos 2x}_{II}\,dx\right]$$

$\Rightarrow$ On solving $\pi(2\pi^2 - 12)$

$\Rightarrow \alpha = 2, \beta = -12$

$\Rightarrow (\alpha + \beta)^2 = 100$

34. Let f be a real valued continuous function defined on the positive real axis such that

$g(x) = \int_0^x tf(t)dt$. If $g(x^3) = x^6 + x^7$, then value of $\sum_{r=1}^{15} f(r^3)$ is :

माना कि f धनात्मक वास्तविक अक्ष पर परिभाषित एक वास्तविक मानों का संतत फलन इस प्रकार है कि $g(x) = \int_0^x tf(t)dt$ है। यदि $g(x^3) = x^6 + x^7$ है, तो $\sum_{r=1}^{15} f(r^3)$ का मान है :

(A) 340 (B) 320 (C) 270 (D) 310

Ans. D

Sol. $g(x) = \int_0^x tf(t)dt$

$g'(x) = x \cdot f(x)$

$g(x^3) = x^6 + x^7$

$g'(x^3) \cdot 3x^2 = 6x^5 + 7x^6$

$x^3 \cdot f(x^3) \cdot 3x^2 = 6x^5 + 7x^6$

$3 \cdot f(x^3) = 6 + 7x$

$$f(x^3) = \left(\frac{6+7x}{3}\right)$$

$$\sum_{r=1}^{15} f(r^3) = \sum_{r=1}^{15}\left(\frac{6+7r}{3}\right)$$

$$\Rightarrow 2\left(\sum_{r=1}^{15} 1\right) + \left(\frac{7}{3}\right)\left(\sum_{r=1}^{15} r\right)$$

$$\Rightarrow 2(15) + \left(\frac{7}{3}\right)\left(\frac{(15)(16)}{2}\right)$$

$\Rightarrow 30 + 40(7)$

$\Rightarrow 30 + 280 = 310$

35. Let $f: R \to R$ be a twice differentiable function such that $f(2) = 1$. If $F(x) = x\,f(x)$ for all $x \in R$, $\int_0^2 x\,F'(x)\,dx = 6$ and $\int_0^2 x^2 F''(x)\,dx = 40$, then $F'(2) + \int_0^2 F(x)\,dx$ is equal to :

माना कि $f: R \to R$, एक दो बार अवकलनीय फलन इस प्रकार है कि $f(2) = 1$ है। यदि सभी $x \in R$ के लिए, $F(x) = x\,f(x)$, $\int_0^2 x\,F'(x)\,dx = 6$ और $\int_0^2 x^2 F''(x)\,dx = 40$ है, तब $F'(2) + \int_0^2 F(x)\,dx$ बराबर है :

(A) 11 (B) 15 (C) 13 (D) 9

Ans. A

Sol. $f(2) = 1$

$F(x) = xf(x)$ (Given)

$$\int_0^2 \underset{II}{x}\cdot\underbrace{F'(x)}_{I}\,dx = 6$$

Using by part

$$\left[\left(F'(x)\cdot\frac{x^2}{2}\right) - \int F''(x)\cdot\frac{x^2}{2}\right]_0^2 = 6$$

$$\Rightarrow \left[F'(x)\cdot\frac{x^2}{2}\right]_0^2 - \frac{\int_0^2 x^2\cdot F''(x)}{2} = 6$$

$$\Rightarrow F'(2)\cdot(2) - \left(\frac{40}{2}\right) = 6$$

$\Rightarrow F'(2) = 13$

$$\Rightarrow F'(2) + \int_0^2 F(x)\,dx$$

$$\Rightarrow F'(2) + \left[F(x)\cdot x\right]_0^2 - \int_0^2 F'(x)\cdot x\,dx$$

$\Rightarrow F'(2) + 2\cdot F(2) - 6$
$\Rightarrow 13 + 2\cdot(2)\cdot f(2) - 6$
$\Rightarrow 13 + (2)\,(2)\,(1) - 6$
$\Rightarrow 11$

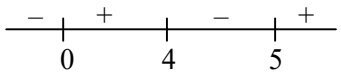
36. Let $f(x)=\int_0^x t\left(t^2-9t+20\right)dt, 1\le x\le 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha+\beta)$ equals :

माना कि $f(x)=\int_0^x t\left(t^2-9t+20\right)dt, 1\le x\le 5$ है। यदि f का परिसर $[\alpha, \beta]$, है, तो $4(\alpha+\beta)$ बराबर है :

(A) 157 (B) 253 (C) 154 (D) 125

Ans. A

Sol. $f'(x) = x^3 - 9x^2 + 20x = x(x-4)(x-5)$



$$\therefore f(x)=\frac{x^4}{4}-\frac{9x^3}{3}+\frac{20x^2}{2}$$

$$f(1)=\frac{1}{4}-3+10=\frac{29}{4}=\alpha$$

$$f(4)=\frac{256}{4}-3(64)$$

$$f(4)=\frac{256}{4}-3(64)+10(16)=32=\beta$$

$$4(\alpha+\beta)=4\left(\frac{29}{4}+32\right)=157$$

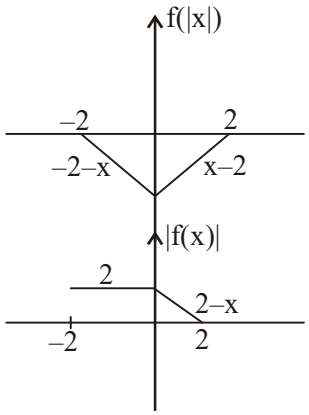
37. Let $f(x)=\begin{cases}-2, & -2\le x\le 0\\ x-2, & 0<x\le 2\end{cases}$ and $h(x)=f(|x|)+|f(x)|$. Then $\int_{-2}^{2} h(x)dx$ is equal to :

माना $f(x)=\begin{cases}-2, & -2\le x\le 0\\ x-2, & 0<x\le 2\end{cases}$ तथा $h(x)=f(|x|)+|f(x)|$ हैं। तो $\int_{-2}^{2} h(x)dx$ बराबर है :

(A) 2 (B) 6 (C) 4 (D) 1

Ans. A

Sol.



$$\therefore h(x)=\begin{cases}-x & x\in[-2,0]\\ 0 & x\in[0,2]\end{cases}$$

$$\int_{-2}^{2} h(x)dx=\int_{-2}^{0}(-x)dx$$

$$=\left(-\frac{x^2}{22}\right)_{-2}^{0}$$

$$=-(-2)=2$$

38. If the value of the integral $\int_{-1}^{1}\frac{\cos\alpha x}{1+3^x}dx$ is $\frac{2}{\pi}$. Then, a value of α is :

यदि समाकलन $\int_{-1}^{1}\frac{\cos\alpha x}{1+3^x}dx$ का मान $\frac{2}{\pi}$ है। तो α का एक मान है :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Ans. C

Sol. Let $I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^x}dx$ _____(1)

$$I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^{-x}}dx$$

(using $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$) _____(2)

Add (1) and (2)

$2I = \int_{-1}^{+1} \cos(\alpha x)dx = 2\int_{0}^{1} \cos(\alpha x)dx$

$I = \frac{\sin\alpha}{\alpha} = \frac{2}{\pi}$ (given)

$\therefore \alpha = \frac{\pi}{2}$

39. Let $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right)dt$, $x \in R$. Then, $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$ is equal to :

माना $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right)dt$, $x \in R$ है। तो $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$ बराबर है :

(A) $\frac{1}{6}$ (B) $-\frac{1}{6}$ (C) $-\frac{2}{3}$ (D) $\frac{2}{3}$

Ans. B

Sol. $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$

Using L Hopital Rule.

$\lim\limits_{x\to 0} \frac{f'(x)}{3x^2} = \lim\limits_{x\to 0} \frac{x + \sin\left(1 - e^x\right)}{3x^2}$ (Again L hospital)

Using L Hopital Rule

$\lim\limits_{x\to 0} \frac{-\left[\sin\left(1 - e^x\right)\left(-e^x\right)\cdot e^x + \cos\left(1 - e^x\right)\cdot e^x\right]}{6}$

$= -\frac{1}{6}$

40. The integral $\int_0^{\frac{\pi}{4}} \frac{136\sin x}{3\sin x + 5\cos x}dx$ is equal to :

समाकलन $\int_0^{\frac{\pi}{4}} \frac{136\sin x}{3\sin x + 5\cos x}dx$ बराबर है :

(A) $3\pi - 30\log_e 2 + 20\log_e 5$ (B) $3\pi - 50\log_e 2 + 20\log_e 5$

(C) $3\pi - 10\log_e\left(2\sqrt{2}\right) + 10\log_e 5$ (D) $3\pi - 25\log_e 2 + 10\log_e 5$

Ans. B

Sol. $(136)\int_0^{\pi/4} \frac{\sin x}{3\sin x + 5\cos x}dx$

$136\sin x = \lambda(3\sin x + 5\cos x) + \mu(3\cos x - 5\sin x)$

Compare coffi of sinx, cosx

$136 = 3\lambda - 5\mu$ ...(1)

$0 = 5\lambda + 3\mu$ ...(2)

$408 = 34\lambda \Rightarrow \lambda = 12$

$\mu = -20$

$(136)\int_0^{\pi/4} \frac{\sin x}{3\sin x + 5\cos x}dx$

$= 12(x)_0^{\pi/4} - 20\left[\ell n(3\sin x + 5\cos x)\right]_0^{\pi/4}$

$= 12\frac{\pi}{4} - 20\left[\ell n 4\sqrt{2} - \ell n 5\right]$

$3\pi - 20\ell n\left(4\sqrt{2}\right) + 20\ell n 5$

$= 3\pi - 50\ell n 2 + 20\ell n 5$

41. Let $\beta(m,n)=\int_0^1 x^{m-1}(1-x)^{n-1}\,dx,\ m,n>0$. If $\int_0^1\left(1-x^{10}\right)^{20}dx=a\times\beta(b,c)$, then 100 (a + b + c) equals :

माना $\beta(m,n)=\int_0^1 x^{m-1}(1-x)^{n-1}\,dx,\ m,n>0$ है। यदि $\int_0^1\left(1-x^{10}\right)^{20}dx=a\times\beta(b,c)$ है, तो 100 (a + b + c) बराबर है :

(A) 1021 (B) 2120 (C) 2012 (D) 1120

Ans. B

Sol. $I=\int_0^1\left(1-x^{10}\right)^{20}dx$

$x^{10}=t\Rightarrow x=(t)^{1/10}$

$I=\frac{1}{10}\int_0^1(t)^{-9/10}(1-t)^{20}\,dt$

$a=\frac{1}{10},\ b=\frac{1}{10},\ c=21$

$100\left(\frac{1}{10}+\frac{1}{10}+21\right)$

$\Rightarrow 10+10+2100=2120$

42. $\int_0^{\pi/4}\frac{\cos^2x\sin^2x}{\left(\cos^3x+\sin^3x\right)^2}dx$ is equal to :

$\int_0^{\pi/4}\frac{\cos^2x\sin^2x}{\left(\cos^3x+\sin^3x\right)^2}dx$ बराबर है :

(A) 1/9 (B) 1/6 (C) 1/3 (D) 1/12

Ans. B

Sol. $\int_0^{\pi/4}\frac{\cos^2x\sin^2x}{\left(\cos^3x+\sin^3x\right)^2}dx$

$\Rightarrow\ \int_0^{\pi/4}\frac{\tan^2x\cdot\sec^2x}{\left(1+\tan^3x\right)^2}\cdot dx$

Let $1+\tan^3x=t$

$3\cdot\tan^2x\cdot\sec^2x\cdot dx=dt$

$$\Rightarrow \frac{1}{3}\int_1^2 \frac{dt}{\left(t^2\right)^2}$$

$$\Rightarrow \frac{1}{3}\left[-\frac{1}{t}\right]_1^2$$

$$\Rightarrow -\frac{1}{3}\left[+\frac{1}{2}-1\right] \Rightarrow \frac{1}{6}$$

43. The value of $k \in N$ for which the integral $I_n = \int_0^1 \left(1-x^k\right)^n dx$, $n \in N$, satisfies $147\, I_{20} = 148\, I_{21}$ is :

$k \in N$ का मान जिसके लिए समाकलन $I_n = \int_0^1 \left(1-x^k\right)^n dx$, $n \in N$ एवं $147\, I_{20} = 148\, I_{21}$ को संतुष्ट करता है :

(A) 8 (B) 7 (C) 10 (D) 14

Ans. B

Sol. $I_n = \int_0^1 \left(1-x^k\right)^n dx$

$$I_{21} = \int_0^1 \left(1-x^k\right)^{21} dx$$

$$\Rightarrow \int_0^1 \left(1-x^k\right)\left(1-x^k\right)^{20} dx$$

$$\Rightarrow \int_0^1 \left(1-x^k\right)^{20} dx - \int_0^1 x^k\left(1-x^k\right)^{20} dx$$

$$I_{21} = I_{20} - \int_0^1 x^k\left(1-x^k\right)^{20} dx$$

$$\int_0^1 xx^{(k-1)}\left(1-x^k\right)^{20} dx$$

$1 - x^k = t$

$-K(x^{K-1}\, dx) = dt$

Integration by part

$$x\left[-\frac{\left(1-x^K\right)^{21}}{21K}\right]_0^1-\frac{1}{21K}\int_0^1\left(1-x^k\right)^{21}dx$$

$$I_{21}=I_{20}+0-\frac{I_{21}}{21K}$$

$$=\left(\frac{21K+1}{21K}\right)I_{21}=I_{20}$$

$$=(21\ K+1)\ I_{21}=21KI_{20}$$

$$K=7$$

44. Let $\int_\alpha^{\log_e 4}\frac{dx}{\sqrt{e^x-1}}=\frac{\pi}{6}$. Then e^α and $e^{-\alpha}$ are the roots of the equation :

माना $\int_\alpha^{\log_e 4}\frac{dx}{\sqrt{e^x-1}}=\frac{\pi}{6}$ है। तो e^α तथा $e^{-\alpha}$ निम्न में से किस समीकरण के मूल हैं :

(A) $2x^2-5x-2=0$ (B) $x^2-2x-8=0$ (C) $2x^2-5x+2=0$ (D) $x^2+2x-8=0$

Ans. C

Sol. $\int_\alpha^{\log_e 4}\frac{dx}{\sqrt{e^x-1}}=\frac{\pi}{6}$

Let $e^x-1=t^2$

$e^x\,dx=2tdt$

$$\int\frac{2tdt}{e^x\cdot t}=\int\frac{2dt}{t^2+1}$$

$$=2\tan^{-1}t$$

$$=2\tan^{-1}\left(\sqrt{e^x-1}\right)_\alpha^{\log_e 4}$$

$$=2\left[\tan^{-1}\sqrt{3}-\tan^{-1}\sqrt{e^\alpha-1}\right]=\frac{\pi}{6}$$

$$=\frac{\pi}{3}-\tan^{-1}\sqrt{e^\alpha-1}=\frac{\pi}{12}$$

$= \tan^{-1}\sqrt{e^{\alpha}-1} = \frac{\pi}{4}$

$e^{\alpha} = 2, \qquad e^{-\alpha} = \frac{1}{2}$

$x^2 - \left(2+\frac{1}{2}\right)x + 1 = 0$

$2x^2 - 5x + 2 = 0$

45. $\lim\limits_{x\to\frac{\pi}{2}} \left(\dfrac{\int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \left(\sin\left(2t^{\frac{1}{3}}\right) + \cos\left(t^{\frac{1}{3}}\right)\right) dt}{\left(x-\frac{\pi}{2}\right)^2} \right)$ is equal to :

$\lim\limits_{x\to\frac{\pi}{2}} \left(\dfrac{\int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \left(\sin\left(2t^{\frac{1}{3}}\right) + \cos\left(t^{\frac{1}{3}}\right)\right) dt}{\left(x-\frac{\pi}{2}\right)^2} \right)$ बराबर है :

(A) $\frac{11\pi^2}{10}$ (B) $\frac{3\pi^2}{2}$ (C) $\frac{5\pi^2}{9}$ (D) $\frac{9\pi^2}{8}$

Ans. D

Sol. L. hopital's rule

$\Rightarrow \lim\limits_{x\to\frac{\pi}{2}} \dfrac{-3x^2\cdot(\sin 2x + \cos x)}{2\left(x-\frac{\pi}{2}\right)} = \lim\limits_{x\to\frac{\pi}{2}} \dfrac{-3x^2\cos x(2\sin x+1)}{2\left(x-\frac{\pi}{2}\right)}$

$$= \lim_{x\to\frac{\pi}{2}} \frac{-9\times\frac{\pi^2}{4}}{2} \times \frac{\cos x}{x-\frac{\pi}{2}} = \lim_{x\to\frac{\pi}{2}} \frac{-9\pi^2}{8} \times \frac{-\sin x}{1}$$

$$= \frac{9\pi^2}{8}$$

46. The integral $\int_{\frac{1}{4}}^{\frac{3}{4}} \cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ is equal to :

समाकलन $\int_{\frac{1}{4}}^{\frac{3}{4}} \cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ बराबर है :

(A) $\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $-\frac{1}{4}$ (D) $\frac{1}{2}$

Ans. C

Sol. $\int_{\frac{1}{4}}^{\frac{3}{4}} \cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$

$x = \cos 2\theta$

$\Rightarrow dx = (-2\sin 2\theta\, d\theta)$

Take limit as α and β

$$-2\int_{\alpha}^{\beta} \cos 2\theta \cdot \sin 2\theta d\theta$$

$$= \int_{\alpha}^{\beta} \sin 4\theta d\theta$$

$$= \frac{-\cos 4\theta}{4}\Big|_{\alpha}^{\beta}$$

$$= -\frac{1}{4}\left(2\cdot\left(x^2\right)-1\right)\Big|_{1/4}^{3/4}$$

$$= -\frac{1}{4}\left(2x^2-1\right)\Big|_{1/4}^{3/4}$$

$$= -\frac{1}{4}\left(\frac{18}{16} - 1 - \frac{2}{16} + 1\right)$$

$$= -\frac{1}{4}$$

47. The value of the integral $\int_{-1}^{2} \log_e\left(x + \sqrt{x^2+1}\right)dx$ is :

समाकलन $\int_{-1}^{2} \log_e\left(x + \sqrt{x^2+1}\right)dx$ का मान है :

(A) $\sqrt{2} - \sqrt{5} + \log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$ (B) $\sqrt{5} - \sqrt{2} + \log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

(C) $\sqrt{2} - \sqrt{5} + \log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$ (D) $\sqrt{5} - \sqrt{2} + \log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$

Ans. C

Sol. $\int_{-1}^{2} \log_e\left(x + \sqrt{x^2+1}\right)dx$

$$= \left[x\log_e\left(x + \sqrt{x^2+1}\right)\right]_{-1}^{2} - \int_{-1}^{2} \frac{x}{\left(x + \sqrt{x^2+1}\right)}\left(1 + \frac{x}{\sqrt{x^2+1}}\right)dx$$

$$= 2\log_e(2+\sqrt{5}) + \log_e(\sqrt{2}-1) - \int_{-1}^{2} \frac{x}{\sqrt{x^2+1}}dx$$

$$= \log_e\left[(2+\sqrt{5})^2(\sqrt{2}-1)\right] - \left[\sqrt{x^2+1}\right]_{-1}^{2}$$

$$= \log_e\left[\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right] - \sqrt{5} + \sqrt{2}$$

$$= \sqrt{2} - \sqrt{5} + \log_e\left[\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right]$$

48. If (a, b) be the orthocentre of the triangle whose vertices are (1,2), (2,3) and (3,1), and $I_1 = \int_a^b x\sin\left(4x - x^2\right)dx$,

$I_2 = \int_a^b \sin\left(4x - x^2\right)dx$, then $36\frac{I_1}{I_2}$ is equal to :

यदि शीर्षों (1,2), (2,3) तथा (3,1) के त्रिभुज का लंबकेन्द्र (a, b) है, तथा $I_1 = \int_a^b x\sin\left(4x - x^2\right)dx$, $I_2 = \int_a^b \sin\left(4x - x^2\right)dx$ है,

तो $36\frac{I_1}{I_2}$ बराबर है :

(A) 72 (B) 88 (C) 66 (D) 80

Ans. A

Sol. Slope of AB = 1.

Altitude through C

$y - 1 = -1(x - 3)$

$\Rightarrow x + y = 4.$

Orthocentre (a, b) lies on this

$\Rightarrow a + b = 4$ _____(i)

Now, $I_1 = \int_a^b x.\sin\left(4x - x^2\right)dx$

$= \int_a^b (a+b-x)\sin\left(4(a+b-x) - (a+b-x)^2\right)dx$

$= \int_a^b (4-x)\sin\left(4x - x^2\right)dx$ {using (i)}

$\Rightarrow I_1 = 4\cdot I_2 - I_1$

$\Rightarrow \frac{I_1}{I_2} = 2$

$\Rightarrow 36\frac{I_1}{I_2} = 72$

49. For $0 < a < 1$, the value of the integral $\int_0^{\pi} \frac{dx}{1 - 2a\cos x + a^2}$ is :

$0 < a < 1$ के लिए, समाकलन $\int_0^{\pi} \frac{dx}{1 - 2a\cos x + a^2}$ का मान है :

(A) $\frac{\pi}{1-a^2}$ (B) $\frac{\pi^2}{\pi+a^2}$ (C) $\frac{\pi^2}{\pi-a^2}$ (D) $\frac{\pi}{1+a^2}$

Ans. A

Sol. $I=\int_0^{\pi}\frac{dx}{1-2a\cos x+a^2};0<a<1$

$$I=\int_0^{\pi}\frac{dx}{1+2a\cos x+a^2}$$

$$2I=2\int_0^{\pi/2}\frac{2\left(1+a^2\right)}{\left(1+a^2\right)^2-4a^2\cos^2 x}dx$$

$$\Rightarrow I=\int_0^{\pi/2}\frac{2\left(1+a^2\right)\cdot\sec^2 x}{\left(1+a^2\right)^2\cdot\sec^2 x-4a^2}dx$$

$$\Rightarrow I=\int_0^{\pi/2}\frac{2\cdot\left(1+a^2\right)\cdot\sec^2 x}{\left(1+a^2\right)^2\cdot\tan^2 x+\left(1-a^2\right)^2}dx$$

$$\Rightarrow I=\int_0^{\pi/2}\frac{\frac{2\cdot\sec^2 x}{1+a^2}\cdot dx}{\tan^2 x+\left(\frac{1-a^2}{1+a^2}\right)^2}$$

$$\Rightarrow I=\frac{2}{\left(1-a^2\right)}\left[\frac{\pi}{2}-0\right]$$

$$I=\frac{\pi}{1-a^2}$$

50. If the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx=\frac{\pi}{4}(\pi+a)-2$, then the value of a is :

यदि समाकलन $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx=\frac{\pi}{4}(\pi+a)-2$ है, तो a का मान है :

(A) $\frac{3}{2}$ (B) 3 (C) 2 (D) $-\frac{3}{2}$

Ans. B

Sol. By even odd property

$$\int_0^{\frac{\pi}{2}}\left(x^2\cos x+\sin^2 x+1\right)dx$$

$$\int_0^{\frac{\pi}{2}}\left(x^2\cos x+\frac{3}{2}-\frac{\cos 2x}{2}\right)dx$$

$$\left(x^2\sin x-\int 2x\sin x dx+\frac{3x}{2}-\frac{\sin 2x}{4}\right)_0^{\frac{\pi}{2}}$$

$$=\left[x^2\sin x-\left[2x(-\cos x)-\int 2(-\cos x)dx\right]+\frac{3x}{2}-\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$$

$$=\left[x^2\sin x+2x\cos x-2\sin x+\frac{3x}{2}-\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$$

$$=\left[\frac{\pi^2}{4}+0-2+\frac{3\pi}{4}-0\right]-[0]=\frac{\pi^2}{4}+\frac{3\pi}{4}-2$$

51. $\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1}{\left(x-\frac{\pi}{2}\right)^2}\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{t}{3}}\right)dt\right)$ is equal to :

$\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1}{\left(x-\frac{\pi}{2}\right)^2}\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{t}{3}}\right)dt\right)$ का मान है :

(A) $\frac{3\pi}{4}$ (B) $\frac{3\pi^2}{8}$ (C) $\frac{3\pi^2}{4}$ (D) $\frac{3\pi}{8}$

Ans. B

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{1}{3}}\right)}{\left(x-\dfrac{p}{2}\right)^2}=\dfrac{0}{0}$ form

Apply leibniz

$$\lim_{x\to\frac{\pi}{2}}\frac{\cos\left(\frac{\pi}{2}\right)\times 0-\cos(x)\cdot 3x^2}{2\left(x-\frac{\pi}{2}\right)}$$

$$x\to\frac{\pi}{2}+t$$

$$\Rightarrow t\to 0$$

$$\lim_{t\to 0}\frac{-\cos\left(\frac{\pi}{2}+t\right)3\left(\frac{\pi}{2}\right)^2}{2t}$$

$$\lim_{t\to 0}\frac{3\pi^2\sin t}{8t}=\frac{3\pi^2}{8}$$

52. The value of $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n^3}{\left(n^2+k^2\right)\left(n^2+3k^2\right)}$ is :

$\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n^3}{\left(n^2+k^2\right)\left(n^2+3k^2\right)}$ का मान है :

(A) $\dfrac{\left(2\sqrt{3}+3\right)\pi}{24}$ (B) $\dfrac{\pi}{8\left(2\sqrt{3}+3\right)}$ (C) $\dfrac{13\left(2\sqrt{3}-3\right)\pi}{8}$ (D) $\dfrac{13\pi}{8\left(4\sqrt{3}+3\right)}$

Ans. D

Sol. $\lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{k=1}^{n}\dfrac{n^3}{n^4\left(1+\dfrac{k^2}{n^2}\right)\left(1+\dfrac{3k^2}{n^2}\right)}$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{n^3}{\left(1+\frac{k^2}{n^2}\right)\left(1+\frac{3k^2}{n^2}\right)}$$

$$=\int_0^1\frac{dx}{3\left(1+x^2\right)\left(\frac{1}{3}+x^2\right)}$$

$$=\frac{1}{3}\times\frac{3}{2}\int_0^1\frac{\left(x^2-1\right)-\left(x^2+\frac{1}{3}\right)}{\left(1+x^2\right)\left(\frac{1}{3}+x^2\right)}$$

$$=\frac{1}{2}\int_0^1\left[\frac{1}{x^2+\left(\frac{1}{\sqrt{3}}\right)^2}-\frac{1}{1+x^2}\right]dx$$

$$=\frac{1}{2}\left[\sqrt{3}\tan^{-1}\left(\sqrt{3}x\right)\right]_0^1-\frac{1}{2}\left(\tan^{-1}x\right)_0^1$$

$$=\frac{\sqrt{3}}{2}\left(\frac{\pi}{3}\right)-\frac{1}{2}\left(\frac{\pi}{4}\right)$$

$$=\frac{\pi}{2\sqrt{3}}-\frac{\pi}{8}$$

$$=\frac{\pi}{2}\left(\frac{1}{\sqrt{3}}-\frac{1}{4}\right)$$

$$=\frac{\pi}{2}\left(\frac{4-\sqrt{3}}{4\sqrt{3}}\right)\left(\frac{4+\sqrt{3}}{4+\sqrt{3}}\right)$$

$$=\frac{\pi}{2}\left(\frac{16-3}{4\sqrt{3}\left(4+\sqrt{3}\right)}\right)$$

$$=\frac{13\pi}{8\left(4\sqrt{3}+3\right)}$$

53. Let $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to R$ be a differentiable function such that $f(0)=\frac{1}{2}$. If the $\lim\limits_{x\to0}\frac{x\int_0^x f(t)dt}{e^{x^2}-1}=\alpha$, then $8\alpha^2$ is equal to :

माना $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to R$ एक अवकलनीय फलन है तथा $f(0)=\frac{1}{2}$ है। यदि $\lim\limits_{x\to0}\frac{x\int_0^x f(t)dt}{e^{x^2}-1}=\alpha$ है, तो $8\alpha^2$ बराबर है :

(A) 16 (B) 2 (C) 4 (D) 1

Ans. B

Sol. $$\lim_{x\to0}\frac{\int_0^x f(t)dt+xf(x)}{e^{x^2}(2x)}=\lim_{x\to0}\frac{\int_0^x f(t)dt}{e^{x^2}(2x)}+\lim_{x\to0}\frac{f(x)}{e^{x^2}(2)}$$

$$=\lim_{x\to0}\frac{f(x)}{\left(e^{x^2}+xe^{x^2}(2x)\right)}+\frac{1}{4}=\frac{1}{2}$$

$$\Rightarrow 8\alpha^2=2$$

54. Let $f:R\to R$ be defined as $f(x)=ae^{2x}+be^x+cx$. If $f(0)=-1, f'(\log_e 2)=21$ and $\int_0^{\log_e 4}(f(x)-cx)dx=\frac{39}{2}$, then the value of $|a+b+c|$ equals :

माना Let $f:R\to R$, $f(x)=ae^{2x}+be^x+cx$ द्वारा परिभाषित है। यदि $f(0)=-1, f'(\log_e 2)=21$ तथा $\int_0^{\log_e 4}(f(x)-cx)dx=\frac{39}{2}$ हैं, तो $|a+b+c|$ का मान बराबर है :

(A) 12 (B) 16 (C) 8 (D) 10

Ans. C

Sol. $f(x)=ae^{2x}+be^x+cx$ $\qquad f(0)=-1$

$\qquad\qquad a+b=-1$

$f'(x)=2ae^{2x}+be^x+c$ $\qquad f'(\ln2)=21$

$\qquad\qquad 8a+2b+c=21$

$$\int_0^{\ell n4}\left(ae^{2x}+be^x\right)dx=\frac{39}{2}$$

$$\left[\frac{ae^{2x}}{2}+be^{x}\right]_{0}^{\ell n4}=\frac{39}{2}\Rightarrow 8a+4b-\frac{a}{2}-b=\frac{39}{2}$$

$15a + 6b = 39$

$15a - 6a - 6 = 39$

$9a = 45 \Rightarrow a = 5$

$b = -6$

$c = 21 - 40 + 12 = -7$

$a + b + c - 8$

$|a + b + c| = 8$

55. Let a and b be real constants such that the function f defined by $f(x)=\begin{cases}x^2+3x+a & ,x\le 1\\ bx+2 & ,x>1\end{cases}$ be differentiable on R. Then, the value of $\int_{-2}^{2} f(x)dx$ equal :

माना वास्तविक अचर a तथा b इस प्रकार हैं कि फलन $f(x)=\begin{cases}x^2+3x+a & ,x\le 1\\ bx+2 & ,x>1\end{cases}$, R पर अवकलनीय है। तो

$\int_{-2}^{2} f(x)dx$ का मान बराबर है :

(A) 21 (B) $\frac{15}{6}$ (C) 17 (D) $\frac{19}{6}$

Ans. C

Sol. f(x) is continuous at x = 1

$f(1^-) = f(1) = f(1^+)$

$a + 4 = b + 2 \Rightarrow b = a + 2$

f(x) is differential at x = 1

$f'(1^-) = f'(1^+)$

$5 = b \Rightarrow a = 3$

Now $\int_{-2}^{2} f(x)dx = \int_{-2}^{1}\left(x^2+3x+3\right)dx + \int_{1}^{2}(5x+2)dx$

$$=\left(\frac{x^3}{3}+\frac{3x^2}{2}+3x\right)_{-2}^{1}+\left(\frac{5x^2}{2}+2x\right)_{1}^{2}=\left(\frac{1}{3}(1+8)+\frac{3}{2}(1-4)+9\right)+\frac{5}{2}(4-1)+2$$

$$=\frac{1}{3}+\frac{3}{2}+3-\left(\frac{-8}{3}+6-6\right)+\left(10+4-\frac{5}{2}-2\right)$$

$$=6+\frac{3}{2}+12-\frac{5}{2}=18-1=17$$

56. Let y = f(x) be a thrice differentiable function in (–5, 5). Let the tangents to the curve y = f(x) at (1, f(1)) and (3, f(3)) make angles $\frac{\pi}{6}$ and $\frac{\pi}{4}$, respectively with positive x-axis. If $27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ where α, β are integers, then the value of $\alpha+\beta$ equals :

माना फलन y = f(x) अंतराल (–5, 5) तीन बार अवकलनीय है। माना वक्र y = f(x) के बिन्दुओं (1, f(1)) तथा (3, f(3)) पर स्पर्श रेखाऐं धनात्मक x-अक्ष से $\frac{\pi}{6}$ तथा $\frac{\pi}{4}$ के कोण बनाती हैं। यदि $27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ हैं, जहाँ α, β पूर्णांक हैं, तो $\alpha+\beta$ का मान बराबर है :

(A) 26 (B) –16 (C) –14 (D) 36

Ans. A

Sol. $y=f(x)\Rightarrow\frac{dy}{dx}=f'(x)$

$$\left.\frac{dy}{dx}\right)_{(1,f(1))}=f'(1)=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}\Rightarrow f'(1)=\frac{1}{\sqrt{3}}$$

$$\left.\frac{dy}{dx}\right)_{(3,f(3))}=f'(3)=\tan\frac{\pi}{4}=1\Rightarrow f'(3)=1$$

$$27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)\,dt=\alpha+\beta\sqrt{3}$$

$$I=\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)\,dt$$

$f'(t)=z\Rightarrow f''(t)\,dt=dz$

$z=f'(3)=1$

$$z = f'(1) = \frac{1}{\sqrt{3}}$$

$$I = \int_{\frac{1}{\sqrt{3}}}^{1} \left(z^2+1\right)dz = \left(\frac{z^3}{3}+z\right)_{\frac{1}{\sqrt{3}}}^{1}$$

$$= \left(\frac{1}{3}+1\right) - \left(\frac{1}{3}\cdot\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{3}}\right)$$

$$= \frac{4}{3} - \frac{10}{9\sqrt{3}} = \frac{4}{3} - \frac{10}{27}\sqrt{3}$$

$$\alpha + \beta\sqrt{3} = 27\left(\frac{4}{3} - \frac{10}{27}\sqrt{3}\right) = 36 - 10\sqrt{3}$$

$\alpha = 36,\ \beta = -10$

$\alpha + \beta = 36 - 10 = 26$

57. Let $f: R \to R$ be a function defined by $f(x) = \frac{x}{\left(1+x^4\right)^{1/4}}$ and $g(x) = f\left(f\left(f\left(f(x)\right)\right)\right)$. Then, $18\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)dx$ equal to :

माना फलन $f: R \to R,\ f(x) = \frac{x}{\left(1+x^4\right)^{1/4}}$ द्वारा परिभाषित है तथा $g(x) = f\left(f\left(f\left(f(x)\right)\right)\right)$ है। तो $18\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)dx$ बराबर है :

(A) 42 (B) 33 (C) 36 (D) 39

Ans. D

Sol. $$f(x) = \frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}$$

$$fof(x) = \frac{f(x)}{\left(1+f(x)^4\right)^{\frac{1}{4}}} = \frac{\frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}}{\left(1+\frac{x^4}{1+x^4}\right)^{\frac{1}{4}}} = \frac{x}{\left(1+2x^4\right)^{\frac{1}{4}}}$$

$$f\left(f\left(f\left(f\left(x\right)\right)\right)\right)=\frac{x}{\left(1+4x^4\right)^{\frac{1}{4}}}$$

$$18\int_0^{\sqrt{2\sqrt{5}}}\frac{x^3}{\left(1+4x^4\right)^{\frac{1}{4}}}dx$$

Let $1 + 4x^4 = t^4$

$16x^3\,dx = 4t^3\,dt$

$$\frac{18}{4}\int_1^3\frac{t^3dt}{t}$$

$$=\frac{9}{2}\left(\frac{t^3}{3}\right)_1^3$$

$$=\frac{3}{2}[26]=39$$

58. Let a be the sum of all coefficients in the expansion of $(1 - 2x + 2x^2)^{2023}\,(3 - 4x^2 + 2x^3)^{2024}$ and $b=\lim_{x\to0}\left(\frac{\int_0^x\frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$. If the equtions $cx^2 + dx + e = 0$ and $2bx^2 + ax + 4 = 0$ have a common root, where c, d, e ∈ R, then d : c : e equals :

माना $(1 - 2x + 2x^2)^{2023}\,(3 - 4x^2 + 2x^3)^{2024}$ के प्रसार में सभी पदों के गुणांक का योग a है तथा $b=\lim_{x\to0}\left(\frac{\int_0^x\frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$ है।

यदि समीकरणों $cx^2 + dx + e = 0$ तथा $2bx^2 + ax + 4 = 0$ का एक उभयनिष्ठ मूल हैं, जहाँ c, d, e ∈ R हैं, तो d : c : e बराबर है :

(A) 1 : 1 : 4 (B) 4 : 1 : 4 (C) 2 : 1 : 4 (D) 1 : 2 : 4

Ans. A

Sol. By putting x = 1

$a = (1 - 2 + 2)^{2023} (3 - 4 + 2)^{2024} = (1)(1) = 1$

$$b = \lim_{x\to 0} \frac{\int_0^x \frac{\ell n(1+t)}{t^{2024}+1} dt}{x^2} \quad \left(\frac{0}{0} \text{ form}\right)$$

By DL hospital Rule

$$\lim_{x\to 0} \frac{\frac{\ell n(1+x)}{x^{2024}+1}}{2x}$$

$$= \frac{1}{2} \cdot \frac{1}{(0+1)} = \frac{1}{2}$$

Now $cx^2 + dx + e = 0$ and $x^2 + x + 4 = 0$ have a common root.

But $D < 0$ for second equation. So both roots are common.

$$\therefore \frac{c}{1} = \frac{d}{1} = \frac{e}{1}$$

$d : c : e = 1 : 1 : 4$

59. Let $f, g : (0, \infty) \to \mathbb{R}$ be two functions defined by $f(x) = \int_{-x}^{x} \left(|t| - t^2\right) e^{-t^2} dt$ and $g(x) = \int_0^{x^2} t^{1/2} e^{-t} dt$. Then the value of $9\left(f\left(\sqrt{\log_e 9}\right) + g\left(\sqrt{\log_e 9}\right)\right)$ is equal to :

माना $f(x) = \int_{-x}^{x} \left(|t| - t^2\right) e^{-t^2} dt$ तथा $g(x) = \int_0^{x^2} t^{1/2} e^{-t} dt$ द्वारा परिभाषित दो फलन $f, g : (0, \infty) \to \mathbb{R}$ हैं। तो $9\left(f\left(\sqrt{\log_e 9}\right) + g\left(\sqrt{\log_e 9}\right)\right)$ का मान बराबर है :

(A) 9 (B) 6 (C) 10 (D) 8

Ans. D

Sol. $f(x) = \int_{-x}^{x} \left(|t| - t^2\right) e^{-t^2} dt$

$\Rightarrow f'(x) = 2 \cdot \left(|x| - x^2\right) e^{-x^2}$ .....(1)

$g(x) = \int_0^{x^2} t^{\frac{1}{2}} e^{-t} dt$

$g'(x) = x^{-x^2}(2x) - 0$

$f'(x) + g'(x) = 2x^{-x^2} - 2x^2 e^{-x^2} + 2x^2 e^{-x^2}$

Integrating both sides w.r.t.x

$$f(x)+g(x)=\int_0^{\alpha}2xe^{-x^2}dx$$

$$x^2=t$$

$$\Rightarrow \int_0^{\sqrt{\alpha}}e^{-t}dt=\left[-e^{-t}\right]_0^{\sqrt{\alpha}}$$

$$=-e^{\left(\log_e(9)^{-1}\right)+1}$$

$$\Rightarrow 9(f(x)+g(x))=\left(1-\frac{1}{9}\right)9=8$$

60. The value of $\int_0^1\left(2x^3-3x^2-x+1\right)^{\frac{1}{3}}dx$ is equal to :

$\int_0^1\left(2x^3-3x^2-x+1\right)^{\frac{1}{3}}dx$ का मान बराबर है :

(A) 1 (B) 2 (C) 0 (D) –1

Ans. C

Sol. $2x^3-3x^2-x+1=\frac{1}{4}(2x-1)\left((2x-1)^2-5\right)$

$$I=\int_0^1\left(\left(\frac{(2x-1)\left((2x-1)^2-5\right)}{4}\right)\right)^{\frac{1}{3}}dx$$

put $2x-1=t$

$$I=\frac{1}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(\frac{t\left(t^2-5\right)}{4}\right)^{\frac{1}{3}}dt$$

$=0$ ($\because$ odd function in t)

61. If $\int_0^{\frac{\pi}{3}} \cos^4 x\,dx = a\pi + b\sqrt{3}$, where a and b are rational numbers, then 9a + 8b is equal to :

यदि $\int_0^{\frac{\pi}{3}} \cos^4 x\,dx = a\pi + b\sqrt{3}$, जहाँ a तथा b परिमेय संख्याएँ है, तो 9a + 8b बराबर होगा :

(A) $\frac{3}{2}$ (B) 2 (C) 1 (D) 3

Ans. B

Sol. $\int_0^{\pi/3} \cos^4 x dx$

$$= \int_0^{\pi/3} \left(\frac{1+\cos 2x}{2}\right)^2 dx$$

$$= \frac{1}{4}\int_0^{\pi/3} \left(1 + 2\cos 2x + \cos^2 2x\right) dx$$

$$= \frac{1}{4}\left[\int_0^{\pi/3} dx + 2\int_0^{\pi/3} \cos 2x dx + \int_0^{\pi/3} \frac{1+\cos 4x}{2} dx\right]$$

$$= \frac{1}{4}\left[\frac{\pi}{3} + (\sin 2x)_0^{\pi/3} + \frac{1}{2}\left(\frac{\pi}{3}\right) + \frac{1}{8}(\sin 4x)_0^{\pi/3}\right]$$

$$= \frac{1}{4}\left[\frac{\pi}{3} + (\sin 2x)_0^{\pi/3} + \frac{1}{2}\left(\frac{\pi}{3}\right) + \frac{1}{8}(\sin 4x)_0^{\pi/3}\right]$$

$$= \frac{1}{4}\left[\frac{\pi}{2} + \frac{\sqrt{3}}{2} + \frac{1}{8}\times\left(-\frac{\sqrt{3}}{2}\right)\right]$$

$$= \frac{\pi}{2} + \frac{7\sqrt{3}}{64}$$

$$\therefore a = \frac{1}{8}; b = \frac{7}{64}$$

$$\therefore 9a + 8b = \frac{9}{8} + \frac{7}{8} = 2$$

62. Let f be a differentiable function such that $x^2 f(x) - x = 4\int_0^x t f(t)dt$, $f(1) = \frac{2}{3}$. Then 18f(3) is equal to :

माना एक अवकलनीय फलन f के लिए $x^2f(x)-x=4\int_0^x tf(t)dt$, $f(1)=\frac{2}{3}$ हैं। तो 18f(3) बराबर है :

(A) 180 (B) 210 (C) 160 (D) 150

Ans. C

Sol. Differentiate the given equation

$\Rightarrow 2xf(x)+x^2f'(x)-1=4xf(x)$

$\Rightarrow x^2\frac{dy}{dx}-2xy=1$

$\Rightarrow \frac{dy}{dx}+\left(-\frac{2}{x}\right)y=\frac{1}{x^2}$

$I.F.=e^{\int-\frac{2}{x}\ell nx}=\frac{1}{x^2}$

$\therefore y\left(\frac{1}{x^2}\right)=\int\frac{1}{x^4}dx$

$\Rightarrow \frac{y}{x^2}=\frac{-1}{3x^3}+c$

$\Rightarrow y=-\frac{1}{3x^3}+c$

$\Rightarrow y=-\frac{1}{3x}+cx^2$

$\because f(1)=\frac{2}{3}=-\frac{1}{3}+c\Rightarrow c=1$

$f(x)=-\frac{1}{3x}+x^2$

$18f(3)=160$

63. Let f be a continuous function satisfying $\int_0^{t^2}\left(f(x)+x^2\right)dx=\frac{4}{3}t^3,\forall t>0$. Then $f\left(\frac{\pi^2}{4}\right)$ is equal to :

माना f एक सतत फलन है तथा $\int_0^{t^2}\left(f(x)+x^2\right)dx=\frac{4}{3}t^3,\forall t>0$ है। तो $f\left(\frac{\pi^2}{4}\right)$ बराबर है :

(A) $\left(1-\pi\left(1+\frac{\pi^3}{16}\right)\right.$ (B) $-\pi\left(1-\frac{\pi^2}{16}\right)$ (C) $\pi^2\left(1-\frac{\pi^3}{16}\right)$ (D) $\pi\left(1-\frac{\pi^3}{16}\right)$

Ans. D

Sol. $\int_0^{t^2}\left(f(x)+x^2\right)dx=\frac{4}{3}t^3,\forall t>0$

$\left(f\left(t^2\right)+t^4\right)=2t$

$f\left(t^2\right)=2t-t^4$

$t=\frac{\pi}{2}\Rightarrow f\left(\frac{\pi^2}{4}\right)=\frac{2\pi}{2}-\frac{\pi^4}{16}$

$=\pi-\frac{\pi^4}{16}=\pi\left(1-\frac{\pi^3}{16}\right)$

64. The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ is equal to :

निम्न $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ का मान ज्ञात कीजिए :

(A) $\log_e\left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$ (B) $\log_e\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$

(C) $\log_e\left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$ (D) $\log_e\left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$

Ans. B

Sol. $I=\int_{-\log_e 2}^{\log_e 2} e^x\left[\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right]dx$

Put $e^x=t$

$e^xdx=dt$

$I=\int_{\frac{1}{2}}^{2}1\times\log_e\left[t+\sqrt{1+t^2}\right]dt$

$=\left[t\ln\left(\sqrt{t^2+1}+t\right)\right]_{\frac{1}{2}}^{2}-\int_{1/2}^{2}\frac{t}{\sqrt{t^2+1}}dt$

$$=\left[t\ln\sqrt{t^2+1}+t-\sqrt{t^2+1}\right]_{\frac{1}{2}}^{2}$$

$$=\left[2\ln\sqrt{5}+2-\sqrt{5}\right]-\left[\frac{1}{2}\ln\left(\sqrt{\frac{5}{2}}+\frac{1}{2}\right)-\sqrt{\frac{5}{2}}\right]$$

$$=\ln\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$$

65. Let the function $f:[0,2]\to\mathbb{R}$ be defined as $f(x)=\begin{cases}e^{\min\{x^2,x-[x]\},} & x\in[0,1)\\ e^{[x-\log_e x]}, & x\in[1,2]\end{cases}$ where [t] denotes the greatest integer less than or equal to t. Then the value of the integral $\int_0^2 xf(x)dx$ is :

माना $f:[0,2]\to\mathbb{R}$, $f(x)=\begin{cases}e^{\min\{x^2,x-[x]\},} & x\in[0,1)\\ e^{[x-\log_e x]}, & x\in[1,2]\end{cases}$ द्वारा परिभाषित है, जहाँ [t] का महत्तम पूर्णांक ≤ t है। तो समाकलन $\int_0^2 xf(x)dx$ का मान है :

(A) $2e-1$ (B) $1+\frac{3e}{2}$ (C) $2e-\frac{1}{2}$ (D) $(e-1)\left(e^2+\frac{1}{2}\right)$

Ans. C

Sol. $f(x)=\begin{cases}e^{x^2} & x\in(0,1)\\ e & x\in[1,2)\end{cases}$

Now $\int_0^2 xf(x)dx=\int_0^1 xe^{x^2}dx+\int_1^2 x\cdot edx=2e-\frac{1}{2}$

66. Let [.] denote the greatest integer function. If $\int_0^{e^3}\left[\frac{1}{e^{x-1}}\right]dx=\alpha-\log_e 2$, then α^3 is equal to _____.

माना [.] महत्तम पूर्णांक फलन है। यदि $\int_0^{e^3}\left[\frac{1}{e^{x-1}}\right]dx=\alpha-\log_e 2$ है तो α^3 बराबर है _____.

Ans. 8

Sol. $f(x)=\frac{1}{e^{x-1}}=e^{1-x}$

$f(x)=2$ $\quad$ $f(x)=1$

$\frac{1}{e^{x-1}}=2$ $\quad$ $x=1$

$x=1-\ell n2$

$f(0)=e^1=2.71$

$f\left(e^3\right)=e^{1-e^3}\in(0,1)$

$I=\int_0^{1-\ell n2}2dx+\int_{1-\ell n2}^{1}1dx+\int_1^{e^3}0dx$

$=(1-\ell n2-0)+1(1-1+\ell n2)+0$

$\alpha=\ell n2=2-\ell n2$

$\alpha=2$

$\alpha^3=8$

67. Let $f:(0,\infty)\to R$ be a twice differentiable function. If for some $a\neq 0,\int_0^1 f(\lambda x)d\lambda=af(x), f(1)=1$ and $f(16)=\frac{1}{8}$, then $16-f'\left(\frac{1}{16}\right)$ is equal to _____.

माना कि $f:(0,\infty)\to R$, दो बार अवकलनीय फलन है। यदि किसी $a\neq 0$ के लिए $\int_0^1 f(\lambda x)d\lambda=af(x), f(1)=1$ और $f(16)=\frac{1}{8}$ है, तो $16-f'\left(\frac{1}{16}\right)$ बराबर है_____.

Ans. 112

Sol. $f:(0,\infty)\to R$

$\int_0^1 f(\lambda x)d\lambda=af(x)$ $\quad$ $f(1)=1$

$\lambda x=t$

$xd\lambda=dt$

$\int_0^x f(t)\frac{dt}{x}=af(x)$

$\int_0^x f(t)dt=axf(x)$

$f(x) = a(x\,f'(x) + f(x))$

$f(x) = axf'(x) + af(x)$

$f(x) = y$

$$f'(x) = \frac{dy}{dx}$$

$$y = ax\frac{dy}{dx} + ay$$

$$(1-a)y = \frac{axdy}{dx}$$

$$\int (1-a)\frac{dx}{x} = \int a\frac{dy}{y}$$

$(1-a)\ell nx = a\ell ny + C$

$(1-a)\ell nx = a\,\ell n\,f(x) + C$

$(1-a) \times 0 = a \times 0 + C$

$C = 0$

$(1-a)\ell nx = a\ell nf(x)$

$$(1-a)\,\ell n16 = a \times \ell n\left(\frac{1}{8}\right)$$

$4(1-a)\ell n2 = -3a\ell n2$

$4\ell n2 - 4a\ell n2 = -3a\ell n2$

$4\ell n2 = a\ell n2$

$a = 4$

$-3\ell nx = 4\ell nf(x)$

$$\ell nf(x) = -\frac{3}{4}\ell nx$$

$$f(x) = e^{\frac{-3}{4}\ell nx}$$

$$f(x) = e^{\ell nx - \frac{3}{4}}$$

$$f(x) = x^{-\frac{3}{4}}$$

$$f'(x) = -\frac{3}{4} \times x^{-\frac{3}{4}-1}$$

$$f'(x) = -\frac{3}{4}x^{-\frac{7}{4}}$$

$$f'\left(\frac{1}{16}\right)=-\frac{3}{4}\times\left(\frac{1}{16}\right)^{\frac{-7}{4}}$$

$$=-\frac{3}{4}\times 2^{-4\times-\frac{7}{4}}$$

$$=-\frac{3}{4}\times 2^{7}$$

$$=-3\times 2^5=-96$$

$$16-f'\left(\frac{1}{16}\right)=16+96=112$$

68. If $\lim\limits_{t\to 0}\left(\int\limits_0^1 (3x+5)^t\,dx\right)^{\frac{1}{t}}=\frac{\alpha}{5e}\left(\frac{8}{5}\right)^{\frac{2}{3}}$, then α is equal to _____.

यदि $\lim\limits_{t\to 0}\left(\int\limits_0^1 (3x+5)^t\,dx\right)^{\frac{1}{t}}=\frac{\alpha}{5e}\left(\frac{8}{5}\right)^{\frac{2}{3}}$ है, तो α बराबर है_____.

Ans. 64

Sol. 1^{∞} from

$$\text{Now } L=e^{\lim\limits_{t\to 0}\frac{1}{t}\left(\frac{(3x+5)^{t+1}}{3(t+1)}\Big|_0^1-1\right)}$$

$$=e^{\lim\limits_{t\to 0}\frac{8^{t+1}-5^{t+1}-3t-3}{3t(t+1)}}$$

$$=e^{\frac{8\ell n8-5\ell n5-3}{3}}$$

$$=\left(\frac{8}{5}\right)^{2/3}\left(\frac{64}{5}\right)=\frac{\alpha}{5e}\left(\frac{8}{5}\right)^{2/3}$$

On comparing

$\alpha = 64$

69. If $24\int\limits_0^{\pi/4}\left(\sin\left|4x-\frac{\pi}{12}\right|+[2\sin x]\right)dx=2\pi+\alpha$, where [.] denotes the greatest integer function, then α is equal to

_____.

यदि $24\int_0^{\pi/4}\left(\sin\left|4x-\frac{\pi}{12}\right|+[2\sin x]\right)dx=2\pi+\alpha$ जहाँ [.] महत्तम पूर्णांक फलन निर्दिष्ट करता है, तो α बराबर है _____.

Ans. 12

Sol. $=24\int_0^{\frac{\pi}{48}}-\sin\left(4x-\frac{\pi}{12}\right)+\int_{\pi/48}^{\pi/4}\sin\left(4x-\frac{\pi}{12}\right)+\int_0^{\pi/6}[0]dx+\int_{\pi/6}^{\pi/4}\left[\underbrace{2\sin x}_{1}\right]dx$

$$=24\left[\frac{\left(1-\cos\frac{\pi}{12}\right)}{4}-\frac{\left(-\cos\frac{\pi}{12}-1\right)}{4}\right]+\frac{\pi}{4}-\frac{\pi}{6}$$

$$=24\left(\frac{1}{2}+\frac{\pi}{12}\right)=2\pi+12$$

$\alpha = 12$

70. If the shortest distance between the lines $\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$ and $\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}k$, and $\int_0^k\left[x^2\right]dx=\alpha-\sqrt{\alpha}$, where [x] denotes the greatest integer function, then $6\alpha^3$ is equal to _____.

यदि रेखाओं $\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$ तथा $\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$ के बीच न्यूनतम $\frac{38}{3\sqrt{5}}k$ है तथा $\int_0^k\left[x^2\right]dx=\alpha-\sqrt{\alpha}$ है, जहाँ [x] महत्तम पूर्णांक फलन है, तो $6\alpha^3$ बराबर है _____

Ans. 48

Sol. $L_1\equiv\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$

$L_2\equiv\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$

Shortest distance between L_1 and L_2 is $=\frac{\begin{vmatrix}5&5&-9\\2&3&4\\1&-3&2\end{vmatrix}}{\sqrt{(18)^2+(0)^2+(-9)^2}}$

$$\frac{38}{3\sqrt{5}}k=\frac{90+0+81}{9\sqrt{5}}=\frac{19}{\sqrt{5}}$$

$$\Rightarrow k=\frac{3}{2}$$

Now $\int_0^k\left[x^2\right]dx=\int_0^{\frac{3}{2}}\left[x^2\right]dx$

$$=\int_0^1\left[x^2\right]dx+\int_1^{\sqrt{2}}\left[x^2\right]dx+\int_{\sqrt{2}}^{\frac{3}{2}}\left[x^2\right]dx$$

$$=0+\left(\sqrt{2}-1\right)+2\left(\frac{3}{2}-\sqrt{2}\right)$$

$$=2-\sqrt{2}$$

$\Rightarrow \alpha = 2$

$\Rightarrow 6\alpha^3 = 48$

71. If $\int_0^{\frac{\pi}{4}}\frac{\sin^2 x}{1+\sin x\cos x}dx=\frac{1}{a}\log_e\left(\frac{a}{3}\right)+\frac{\pi}{b\sqrt{3}}$, where a, b $\in$ N, then a + b is equal to _____.

यदि $\int_0^{\frac{\pi}{4}}\frac{\sin^2 x}{1+\sin x\cos x}dx=\frac{1}{a}\log_e\left(\frac{a}{3}\right)+\frac{\pi}{b\sqrt{3}}$, जहाँ a, b $\in$ N है, तो a + b बराबर है _____

Ans. 8

Sol. $\ell=\int_0^{\frac{\pi}{4}}\frac{\sin^2 x}{1+\sin x\cos x}dx=\int_0^{\frac{\pi}{4}}\frac{\sin^2 x}{\sin^2 x+\cos^2 x+\sin x\cos x}dx$

$$\ell=\int_0^{\frac{\pi}{4}}\frac{\tan^2 x}{1+\tan x+\tan^2 x}dx=\int_0^{\frac{\pi}{4}}\frac{\tan^2 x\cdot\sec^2 x dx}{\left(1+\tan^2 x\right)\left(1+\tan x+\tan^2 x\right)}$$

Let $\tan x = t$

$$\ell=\int_0^1\frac{t^2}{\left(1+t^2\right)\left(1+t+t^2\right)}dt$$

$$=\int_0^1\left(\frac{x}{1+x^2}-\frac{x}{1+x+x^2}\right)dx$$

$$= \frac{1}{2}\int_0^1 \frac{2x}{1+x^2}dx - \int_0^1 \frac{\frac{1}{2}(2x+1)-\frac{1}{2}}{1+x+x^2}dx$$

$$= \frac{1}{2}\ell n2 - \frac{1}{2}\ln 3 + \frac{1}{2}\int_0^1 \frac{dx}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{2}\cdot\frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{2x+1}{\sqrt{3}}\right]_0^1$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{\sqrt{3}}\left(\frac{\pi}{3}-\frac{\pi}{6}\right)$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{\sqrt{3}}\cdot\frac{\pi}{6}$$

$\therefore$ a = 2, b = 6

$\therefore$ a + b = 8

72. If $f(t) = \int_0^{\pi} \frac{2x\,dx}{1-\cos^2 t\sin^2 x}$, $0 < t < \pi$, then the value of $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ equals _____.

यदि $f(t) = \int_0^{\pi} \frac{2x\,dx}{1-\cos^2 t\sin^2 x}$, $0 < t < \pi$ है, तो $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ का मान है _____

Ans. 1

Sol. $f(t) = \int_0^{\pi} \frac{2xdx}{1-\cos^2 t\sin^2 x}$

$x \to \pi - x$

$$f(t) = \int_0^{\pi} \frac{2(\pi-x)dx}{1-\cos^2 t\sin^2 x} = 2\pi\int_0^{\pi} \frac{dx}{1-\cos^2 t\sin^2 x} - f(t)$$

$$\Rightarrow f(t) = \pi\int_0^{\pi} \frac{dx}{1-\cos^2 t\sin^2 x}$$

$$= 2\pi\int_0^{\frac{\pi}{2}} \frac{dx}{1-\cos^2 t\sin^2 x}$$

$$f(t) = 2\pi\int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t \tan^2 x}$$

$$I_1 = \int \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t \tan^2 x}$$

Put cost tanx = $\lambda \Rightarrow$ cost $\sec^2 x dx = d\lambda$

$$I_1 = \int \frac{d\lambda}{\cos t.\left(1+\lambda^2 \sec^2 t - \lambda^2\right)} = \int \frac{d\lambda}{\cos t\left(1+\lambda^2 \tan^2 t\right)}$$

$$= \frac{1}{\cos t.\tan^2 t}.\int \frac{d\lambda}{\lambda^2 + \cos^2 t} = \frac{1}{\cos t \tan^2 t} \times \frac{1}{\cos t} \tan^{-1}\left(\lambda \tan t\right)$$

$$= \frac{1}{\sin t} \tan^{-1}\left(\sin t \tan x\right)$$

$$\Rightarrow f(t) = \frac{2\pi}{\sin t} \tan^{-1}\left(\sin t \tan x\right)\Big]_0^{\frac{\pi}{2}} = \frac{\pi^2}{\sin t}$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\pi^2 dt}{f(t)} = \int_0^{\frac{\pi}{2}} \sin t dt = 1$$

73. Let $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}, k \in \mathbb{N}$. Then the value of $\sum_{k=1}^{10} \frac{1}{7\left(r_k - 1\right)}$ is equal to _____.

माना $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}, k \in \mathbb{N}$ है। तो $\sum_{k=1}^{10} \frac{1}{7\left(r_k - 1\right)}$ का मान बराबर है _____

Ans. 65

Sol. $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}$

$$\int_0^1 \left(1-x^7\right)^{k+1} dx = \int_0^1 \left(1-x^7\right)^k \cdot \left(1-x^7\right) dx$$

$$= \int_0^1 \left(1-x^7\right)^k dx - \int_0^1 x^7 \left(1-x^7\right)^k dx$$

$$= \int_0^1 \left(1-x^7\right)^k - \int_0^1 x \cdot x^6 \left(1-x^7\right)^k dx$$

$$= \int_0^1 \left(1-x^7\right)^k - x\int_0^1 x^6 \left(1-x^7\right)^k dx + \int_0^1 \left[1 \cdot \int x^6 \cdot \left(1-x^7\right)^k dx\right] dx$$

$$= \int_0^1 \left(1-x^7\right)^k - x\frac{\left(1-x^7\right)^{k+1}}{-7(k+1)}\Bigg|_0^1 + \int_0^1 \frac{\left(1-x^7\right)^{k+1}}{-7(k+1)} dx$$

$$\Rightarrow \int_0^1 \left(1-x^7\right)^{k+1} dx \left\{1+\frac{1}{7(k+1)}\right\} = \int_0^1 \left(1-x^7\right)^k dx - 0$$

$$\frac{7k+8}{7(k+1)} = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}$$

$$\sum_{k=1}^{10} \frac{1}{7\left(r_k - 1\right)}$$

$$\Rightarrow \sum_{k=1}^{10} \frac{1}{7\left(1+\frac{1}{7k+7}-1\right)}$$

$$\Rightarrow \sum_{k=1}^{10} (K+1) \qquad \Rightarrow \frac{10\times 11}{2} + 10 \Rightarrow 55+10 = 65$$

74. Let [t] denote the largest integer less than or equal to t. If

$$\int_0^3 \left(\left[x^2\right] + \left[\frac{x^2}{2}\right]\right) dx = a + b\sqrt{2} - \sqrt{3} - \sqrt{5} + c\sqrt{6} - \sqrt{7}$$, where a, b, c $\in$ Z, then a + b + c is equal to _____.

माना [t], t से कम या t के बराबर महत्तम पूर्णांक को दर्शाता है। यदि

$$\int_0^3 \left(\left[x^2\right] + \left[\frac{x^2}{2}\right]\right) dx = a + b\sqrt{2} - \sqrt{3} - \sqrt{5} + c\sqrt{6} - \sqrt{7}$$, जहाँ a, b, c $\in$ Z तो, a + b + c बराबर है _____

Ans. 23

Sol. $\int_0^3 \left(\left[x^2\right] + \int_0^3 \left[\frac{x^2}{2}\right]\right) dx$

$$= \int_0^1 0dx + \int_1^{12} 1dx + \int_{\sqrt{2}}^{\sqrt{3}} 2dx$$

$$+ \int_{\sqrt{3}}^{2} 3dx + \int_2^{\sqrt{5}} 4dx + \int_{\sqrt{5}}^{\sqrt{6}} 5dx + \int_{\sqrt{6}}^{\sqrt{7}} 6dx + \int_{\sqrt{7}}^{\sqrt{8}} 7dx + \int_{\sqrt{8}}^{3} 8dx + \int_0^{\sqrt{2}} 0dx + \int_{\sqrt{2}}^{2} 1dx$$

$$+ \int_2^{\sqrt{6}} 2dx + \int_{\sqrt{6}}^{\sqrt{8}} 3dx + \int_{\sqrt{8}}^{3} 4dx = 31 - 6\sqrt{2} - \sqrt{3} - \sqrt{5} - 2\sqrt{5}\sqrt{6} - \sqrt{7}$$

$a = 31$ $b = -6$, $c = -2$

$a + b + c = 31 - 6 - 2 = 23$

75. Let $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}} - \frac{2n}{\left(n^2+1\right)\sqrt{n^4+1}} + \frac{n}{\sqrt{n^4+16}} - \frac{8}{\left(n^2+4\right)\sqrt{n^4+16}} + ... + \frac{n}{\sqrt{n^4+n^4}} - \frac{2n\cdot n^2}{\left(n^2+n^2\right)\sqrt{n^4+n^4}}\right)$ be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to _____.

माना $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}} - \frac{2n}{\left(n^2+1\right)\sqrt{n^4+1}} + \frac{n}{\sqrt{n^4+16}} - \frac{8}{\left(n^2+4\right)\sqrt{n^4+16}} + ... + \frac{n}{\sqrt{n^4+n^4}} - \frac{2n\cdot n^2}{\left(n^2+n^2\right)\sqrt{n^4+n^4}}\right) = \frac{\pi}{k}$

है, जब प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मान लिए गए है। तो k^2 बराबर है _____

Ans. 32

Sol. $\sum\limits_{r=1}^{\infty} \frac{n}{\sqrt{n^4+r^4}} - \frac{2nr^2}{\left(n^2+r^2\right)\sqrt{n^4+r^4}}$

$$\sum_{r=1}^{\infty}\frac{\frac{1}{n}}{\sqrt{1+\left(\frac{r}{n}\right)^4}}-\frac{2\left(\frac{1}{n}\right)\left(\frac{r}{n}\right)^2}{\left(1+\left(\frac{r}{n}\right)^2\right)\sqrt{1+\left(\frac{r}{n}\right)^4}}$$

$$\Rightarrow\int_0^1\frac{dx}{\sqrt{1+x^4}}-\frac{2x^2dx}{\left(1+x^2\right)\sqrt{1+x^4}}$$

$$\Rightarrow\int_0^1\frac{1-x^2}{\left(1+x^2\right)\sqrt{1+x^4}}dx$$

$$\Rightarrow\int_0^1\frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}dx$$

$$\Rightarrow-\int_0^1\frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx$$

$$x+\frac{1}{x}=t\Rightarrow1-\frac{1}{x^2}dx=dt$$

$$\Rightarrow-\int_{\infty}^{2}\frac{dt}{t\sqrt{t^2-2}}$$

$$\Rightarrow-\int_{\infty}^{2}\frac{tdt}{t^2\sqrt{t^2-2}}$$

take $t^2-2=\alpha^2$

$tdt=ad\alpha$

$$\Rightarrow-\int_{\infty}^{\sqrt{2}}\frac{\alpha d\alpha}{\left(\alpha^2+2\right)\alpha}$$

$$\Rightarrow-\int_{\infty}^{\sqrt{2}}\frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\tan^{-1}\frac{\alpha}{\sqrt{2}}\Big]_{\infty}^{\sqrt{2}}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\{\tan^{-1}\} + \frac{1}{\sqrt{2}}\tan^{-1}\infty$$

$$\Rightarrow \frac{1}{\sqrt{2}}\left\{\frac{\pi}{2}-\frac{\pi}{4}\right\}$$

$$\Rightarrow \frac{\pi}{\sqrt{2}} = \frac{\pi}{K}$$

So, $K = 4\sqrt{2}$

$K^2 = 32$

76. Let $f(x) = \int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$, where g is a continuous odd function.

If $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx = \left(\frac{\pi}{\alpha}\right)^2 - \alpha$, then α is equal to _____.

माना $f(x) = \int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$ है, जहाँ g एक संतत विषम फलन है। यदि

यदि $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx = \left(\frac{\pi}{\alpha}\right)^2 - \alpha$ है, तो α बराबर है _____।

Ans. 2

Sol. $f(x) = \int_0^x g(t)\ell n\left(\frac{1-t}{1+t}\right)dt$

$$f(-x) = \int_0^{-x} g(t)\ell n\left(\frac{1-t}{1+t}\right)dt$$

$$f(-x) = -\int_0^x g(-y)\ell n\left(\frac{1+y}{1-y}\right)dy$$

$$= -\int_0^x g(y)\ell n\left(\frac{1-y}{1+y}\right)dy \text{ (g is odd)}$$

$f(-x) = -f(x) \Rightarrow f$ is also odd

Now,

$$I=\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx$$

$$I=\int_{-\pi/2}^{\pi/2}\left(f(-x)+\frac{x^2e^x\cos x}{1+e^x}\right)dx$$

$$2I=\int_{-\pi/2}^{\pi/2}x^2\cos x\,dx=2\int_0^{\pi/2}x^2\cos x\,dx$$

$$I=\left(x^2\sin x\right)_0^{\pi/2}-\int_0^{\pi/2}2x\sin x\,dx$$

$$=\frac{\pi^2}{4}-2\left(-x\cos x+\int\cos x dx\right)_0^{\pi/2}$$

$$=\frac{\pi^2}{4}-2(0+1)=\frac{\pi^2}{4}-2\Rightarrow\left(\frac{\pi}{2}\right)^2-2$$

$\therefore\ a=2$

77. Let the slope of the line $45x + 5y + 3 = 0$ be $27r_1+\frac{9r_2}{2}$ for some $r_1, r_2 \in R$. Then

$$\lim_{x\to3}\left(\int_3^x\frac{8t^2}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}dt\right)$$ is equal to _____ .

माना किसी $r_1, r_2 \in R$ के लिए रेखा $45x+5y+3=0$ की प्रवणता $27r_1+\frac{9r_2}{2}$ है। तो $\lim_{x\to3}\left(\int_3^x\frac{8t^2}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}dt\right)$

बराबर है _____ .

Ans. 12

Sol. Slope of $45x + 5y + 3 = 0$ is $m = -9$

$$\therefore 27r_1+\frac{9r_2}{2}=-9$$

$$\lim_{x\to3}\left(\int_3^x\frac{8t^2}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}dt\right)$$

$$\lim_{x\to 3}\frac{\int_3^x 8t^2dt}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}$$

$$=\lim_{x\to 3}\frac{8x^2}{\frac{3r_2}{2}-2r_2x-3r_1x^2-3}\ \text{by L.H. rule}$$

$$=\frac{8\times 9}{\frac{3r_2}{2}-6r_2-27r_1-3}$$

$$=\frac{8\times 9}{-\frac{9r_2}{2}-27r_1-3}$$

$$=\frac{8\times 9}{6}=12$$

78. If $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, where α, β and γ are rational numbers, then $3\alpha+4\beta-\gamma$ is equal to _____ .

यदि $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, जहाँ α, β तथा γ परिमेय संख्याऐं हैं, तो $3\alpha+4\beta-\gamma$ बराबर है_____ .

Ans. 6

Sol. $=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}\,dx$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}|\sin x-\cos x|\,dx$$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}(\cos x-\sin x)dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}(\sin x-\cos x)dx$$

$$=-1+2\sqrt{2}-\sqrt{3}$$

$= \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$

$\alpha = -1, \beta = 2, \gamma = -1$

$3\alpha + 4\beta - \gamma = 6$

79. The value of $9\int_0^9 \left[\sqrt{\frac{10x}{x+1}}\right] dx$, where [t] denotes the greatest integer less than or equal to t, is _____.

$9\int_0^9 \left[\sqrt{\frac{10x}{x+1}}\right] dx$, जहाँ [t] महत्तम पूर्णांक $\le$ t है, का मान है _____।

Ans. 155

Sol. $9\int_0^9 \left[\sqrt{\frac{10x}{x+1}}\right] dx$

Let $I = \int_0^9 \left[\sqrt{\frac{10x}{x+1}}\right] dx$

$\frac{10x}{x+1} = 1 \Rightarrow x = \frac{1}{9}$

$\frac{10x}{x+1} = 4 \qquad \Rightarrow x = \frac{2}{3}$

$\frac{10x}{x+1} = 9 \qquad \Rightarrow x = 9$

$I = \int_0^{\frac{1}{9}} 0dx + \int_{\frac{1}{9}}^{\frac{2}{3}} 1dx + \int_{\frac{2}{3}}^{9} 2dx$

$= 0 + [x]_{\frac{1}{9}}^{\frac{2}{3}} + [2x]_{\frac{2}{3}}^{9}$

$I = \frac{2}{3} - \frac{1}{9} + 18 - \frac{4}{3}$

$\Rightarrow I = \frac{155}{9}$

$\Rightarrow 9I = 155$

80. Let $f: R \to R$ be a function defined by $f(x)=\frac{4^x}{4^x+2}$ and $M=\int\limits_{f(a)}^{f(1-a)} x\sin^4\left(x(1-x)\right)dx$,

$N=\int\limits_{f(a)}^{f(1-a)} \sin^4\left(x(1-x)\right)dx$; $a\neq\frac{1}{2}$. If $\alpha M=\beta N$, $\alpha,\beta\in N$, then the least value of $\alpha^2+\beta^2$ is equal to _____.

माना फलन $f: R \to R$, $f(x)=\frac{4^x}{4^x+2}$ द्वारा परिभाषित है तथा $M=\int\limits_{f(a)}^{f(1-a)} x\sin^4\left(x(1-x)\right)dx$,

$N=\int\limits_{f(a)}^{f(1-a)} \sin^4\left(x(1-x)\right)dx$; $a\neq\frac{1}{2}$ हैं। यदि $\alpha M=\beta N$, $\alpha,\beta\in N$ हैं, तो $\alpha^2+\beta^2$ का न्यूनतम मान है _____.

Ans. 5

Sol. $f(x)+f(1-x)=\frac{4^x}{4^x+2}+\frac{4^{1-x}}{4^{1-x}}$

$=\frac{4^x}{4^x+2}+\frac{4}{4+2\cdot4^x}$

$=\frac{4^x+2}{4^x+2}=1$

$\Rightarrow M=\int\limits_{f(a)}^{f(1-a)} x\sin^4\left(x(1-x)\right)dx$

$x\to f(a)+f(1-a)-x=1-x$

$M=\int\limits_{f(a)}^{f(1-a)} (1-x)\sin^4\left((1-x)x\right)dx$

$M=N-M\Rightarrow\frac{M}{N}=\frac{1}{2}=\frac{\beta}{\alpha}$

$\Rightarrow\left(\alpha^2+\beta^2\right)_{least}=1+4=5$

81. If the integral $525\int_0^{\frac{\pi}{2}} \sin 2x \cos^{\frac{11}{2}} x\left(1+\cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx$ is equal to $\left(n\sqrt{2}-64\right)$, then n is equal to _____.

यदि समाकलन $525\int_0^{\frac{\pi}{2}} \sin 2x \cos^{\frac{11}{2}} x\left(1+\cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx = \left(n\sqrt{2}-64\right)$ है, तो n बराबर है_____.

Ans. 176

Sol. $I = 525\int \sin 2x \cos^{\frac{11}{2}} x\left(1+\cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx$

put $\sqrt{\cos x} = t \Rightarrow \cos x = t^2$

$-\sin x\, dx = 2t dt$

$\sin x\, dx = -2t\, dt$

$$I = 525\int_0^1 2(2t)t^2t^{11}\sqrt{1+t^5}\,dt$$

$$= 525\times 4\int_0^1 \sqrt{1+t^5}\cdot t^{14}dt$$

$$= 2100\int_0^1 \sqrt{1+t^5}\cdot t^{14}dt$$

again put $\sqrt{1+t^5} = u \quad \Rightarrow 1+t^5 = u^2$

$5t^4dt = 2udu$

$$I = 2100\int_1^{\sqrt{2}} u\cdot\frac{2u}{5}\cdot\left(u^2-1\right)^2 du$$

$$= 420\times 2\left(\int_1^{\sqrt{2}} u^2\left(u^4-2u^2+1\right)du\right)$$

$$= 840\int_1^{\sqrt{2}} \left(u^6-2u^4+u^2\right)du$$

$$= 840\left[\frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3}\right]_1^{\sqrt{2}}$$

$$= 840\left[\frac{8\sqrt{2}}{7} - \frac{8\sqrt{2}}{5} + \frac{2\sqrt{2}}{3} - \frac{1}{7} + \frac{2}{5} - \frac{1}{3}\right]$$

$$= 176\sqrt{2} - 64$$

Compare with $n\sqrt{2} - 64$

$n = 176$

82. $\left|\frac{120}{\pi^3}\int_0^{\pi}\frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x}dx\right|$ is equal to _____.

$\left|\frac{120}{\pi^3}\int_0^{\pi}\frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x}dx\right|$ बराबर है _____

Ans. 15

Sol. $I = \int_0^{\pi}\frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x}dx$

$$2I = \int_0^{\pi}\frac{\sin x \cos x}{\sin^4 x + \cos^4 x}\left(x^2 - (\pi - x)^2\right)dx$$

$$2I = \int_0^{\pi}\frac{\pi(2x - \pi)\sin x \cos x}{\sin^4 x + \cos^4 x}dx$$

$$\Rightarrow \quad I = \pi\int_0^{\pi/2}\frac{(2x - \pi)\sin x \cos x}{\sin^4 x + \cos^4 x}dx$$

$$\Rightarrow \quad I = \pi\int_0^{\pi/2}\frac{-2x \sin x \cos x}{\sin^4 x + \cos^4 x}dx$$

$$2I = -\pi^2\int_0^{\pi/2}\frac{\sin x \cos x}{\sin^4 x + \cos^4 x}dx$$

$$I = \frac{-\pi}{2 \times 2}\int_0^{\pi/2}\frac{\sin 2x}{\sin^4 x + \cos^4 x}dx$$

$$I = \frac{-\pi^2}{4}\int_0^{\pi/2}\frac{2\tan x \sec^2 x}{\tan^4 x + 1}dx$$

$$I = \frac{-\pi^2}{4}\tan^{-1}\left(\tan^2 x\right)_0^{\pi/2}$$

$$\Rightarrow \quad I = \frac{-\pi^3}{8}$$

$$\therefore \quad \left|\frac{120}{\pi^3}I\right| = 15$$

83. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2}\cos x dx}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)} = \alpha\pi + \beta\log_e\left(3+2\sqrt{2}\right)$, where α, β are integers, then $\alpha^2 + \beta^2$ equals _____.

यदि $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2}\cos x dx}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)} = \alpha\pi + \beta\log_e\left(3+2\sqrt{2}\right)$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha^2 + \beta^2$ बराबर है _____

Ans. 8

Sol. $$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2}\cos x}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)} dx \qquad ...(1)$$

Apply king

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2}\cos x\left(e^{\sin x}\right)}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)} dx \qquad ...(2)$$

adding (1) & (2)

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2}\cos x}{1+\sin^4 x} dx$$

$$I = \int_{0}^{\frac{\pi}{2}} \frac{8\sqrt{2}\cos x}{1+\sin^4 x} dx$$

$\sin x = t$

$$I=\int_0^1 \frac{8\sqrt{2}}{1+t^4}dx$$

$$I=4\sqrt{2}\int_0^1\left(\frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}-\frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}}\right)dt$$

$$I=4\sqrt{2}\int_0^1\frac{\left(1+\frac{1}{t^2}\right)}{\left(t-\frac{1}{t}\right)^2+2}-\frac{\left(1-\frac{1}{t^2}\right)}{\left(t+\frac{1}{t}\right)^2-2}dt$$

Let $t-\frac{1}{t}=z$ & $t+\frac{1}{t}=k$

$$=4\sqrt{2}\left[\int_{-\infty}^0\frac{dz}{z^2+2}-\int_{\infty}^2\frac{dk}{k^2-2}\right]$$

$$=4\sqrt{2}\left[\frac{1}{\sqrt{2}}\tan^{-1}\frac{z}{\sqrt{2}}\right]_{-\infty}^0-\left[\frac{1}{2\sqrt{2}}\ell n\left(\frac{k-\sqrt{2}}{k+\sqrt{2}}\right)\right]_{\infty}^2$$

$$=4\sqrt{2}\left[\frac{\pi}{2\sqrt{2}}-\frac{1}{2\sqrt{2}}\left[\ell n\frac{2-\sqrt{2}}{2+\sqrt{2}}\right]\right]$$

$$=2\pi+2\ell n\left(3+2\sqrt{2}\right)$$

$\alpha = 2$

$\beta = 2$

84. Let $f:(0,\infty)\to R$ and $F(x)=\int_0^x t\,f(t)dt$. If $F(x^2)=x^4+x^5$, then $\sum_{r=1}^{12} f(r^2)$ is equal to _____.

माना $f:(0,\infty)\to R$ तथा $F(x)=\int_0^x t\,f(t)dt$ हैं। यदि $F(x^2)=x^4+x^5$ है, तो $\sum_{r=1}^{12} f(r^2)$ बराबर है _____

Ans. 219

Sol. $F(x)=\int_0^x tf(t)dt$ and $F(x)^2=x^4+x^5$

$F'(x) = xf(x)$ and $2xF'\left(x^2\right) = 4x^3 + 5x^4$

$2F'\left(x^2\right) = 4x^2 + 5x^3$

$2F'(x) = 4x + 5x^{\frac{3}{2}}$

$\Rightarrow 2x + \frac{5}{2}x^{\frac{3}{2}} = xf(x)$

$\Rightarrow f(x) = 2 + \frac{5}{2}x^{\frac{1}{2}}$

$\Rightarrow f\left(x^2\right) = 2 + \frac{5}{2}x$

$\sum_{r=1}^{12} f\left(r^2\right) = \sum_{r=1}^{12}\left(2 + \frac{5}{2}r\right)$

$= 24 + \frac{5}{2}\sum_{r=1}^{12} r$

$= 24 + \frac{5}{2} \times \frac{12 \times 13}{2}$

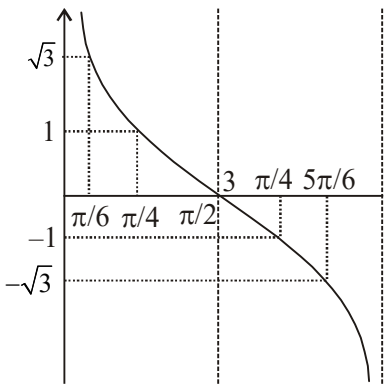
$24 + 195 = 219$

85. Let [t] denote the greatest integer $\le$ t. Then $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}(8[\operatorname{cosec} x] - 5[\cot x])dx$ is equal to _____.

माना [t] महत्तम पूर्णांक $\le$ t है। तो $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}(8[\operatorname{cosec} x] - 5[\cot x])dx$ बराबर है _____

Ans. 14

Sol.



$$I=\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}\left(8[\operatorname{cosec} x]-5[\cot x]\right)dx$$

For $x\in(\pi/6,5\pi/6)$

$\operatorname{cosec} x\in[1,2)$

$[\operatorname{cosec} x]=1$

$$I=\frac{2}{\pi}\left[\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}8(1)dx-\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}5(1)dx-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}5(0)dx-\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}5(-1)dx-\int_{\frac{3\pi}{4}}^{\frac{5\pi}{6}}5(-2)dx\right]=14$$

86. For m, n > 0, let $\alpha(m,n)=\int_0^2 t^m(1+3t)^n\,dt$. If $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$, then p is equal to _____.

m, n > 0 के लिए माना $\alpha(m,n)=\int_0^2 t^m(1+3t)^n\,dt$ है। यदि $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$ तो p बराबर _____ है।

Ans. 32

Sol. $\alpha(m,n)=\int_0^2 t^m(1+3t)^n dt$

If $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$ then P

$$=11\int_0^2 \frac{t^{10}}{II}\frac{(1+3t)^6}{I}+10\int_0^2 t^{11}(1+3t)^5dt$$

$$=11\left[(1+3t)^6\cdot\frac{t^{11}}{11}-\int 6(1+3t)^5\cdot 3\frac{t^{11}}{11}\right]_0^2+18\int_0^2 t^{11}(1+3t)^5dt$$

$$=\left(t^{11}(1+3t)^6\right)_0^2$$

$$=2^{11}(7)^6$$

$$=2^5(14)^6$$

$$=32(14)^6$$

87. Let for $x\in\mathbb{R}, S_0(x)=x, S_k(x)=C_kx+k\int_0^x S_{k-1}(t)dt$, where $C_0=1, C_k=1-\int_0^1 S_{k-1}(x)dx, k=1,2,3,\ldots$ Then $S_2(3)+6C_3$ is equal to _____.

माना $x\in\mathbb{R}, S_0(x)=x, S_k(x)=C_kx+k\int_0^x S_{k-1}(t)dt$ है, जहाँ $C_0=1, C_k=1-\int_0^1 S_{k-1}(x)dx, k=1,2,3,\ldots$ हैं। तो

$S_2(3)+6C_3$ बराबर है _____

Ans. 18

Sol. $C_0 = 1$

$$C_1 = 1-\int_0^1 S_0(x)dx = 1-\int_0^1 xdx = 1-\frac{1}{2}=\frac{1}{2}$$

$$S_1(x) = C_1x+\int_0^x tdt$$

$$=\frac{1}{2}x+\frac{x^2}{2}=\frac{x+x^2}{2}$$

$$C_2 = 1-\int_0^1\left(\frac{x+x^2}{2}\right)dx = 1-\frac{1}{2}\left(\frac{1}{2}+\frac{1}{3}\right)=1-\frac{5}{12}=\frac{7}{12}$$

$$S_2(x) = C_2x+2\int_0^x\left(\frac{t+t^2}{2}\right)dt=\frac{7}{12}x+\left(\frac{x^2}{2}+\frac{x^3}{3}\right)=\frac{7x+6x^2+4x^3}{12}$$

$$S_2(3)=\frac{21+54+108}{12}=\frac{183}{12}=\frac{61}{4}$$

$$C_3 = 1-\int_0^1\left(\frac{7x+6x^2+4x^3}{12}\right)dx$$

$$=1-\frac{1}{12}\left[\frac{7}{2}+\frac{6}{3}+\frac{4}{4}\right]$$

$$=1-\frac{1}{12}\left(\frac{7}{2}+3\right)=\frac{24-7-6}{24}=\frac{11}{24}$$

Now, $S_2(3)+6C_3=\frac{61}{4}+\frac{11}{4}=\frac{72}{4}=18$

88. Let $f_n=\int_0^{\frac{\pi}{2}}\left(\sum_{k=1}^{n}\sin^{k-1}x\right)\left(\sum_{k=1}^{n}(2k-1)\sin^{k-1}x\right)\cos x dx, n\in\mathbb{N}$. Then $f_{21}-f_{20}$ is equal to _____.

माना $f_n=\int_0^{\frac{\pi}{2}}\left(\sum_{k=1}^{n}\sin^{k-1}x\right)\left(\sum_{k=1}^{n}(2k-1)\sin^{k-1}x\right)\cos x dx, n\in\mathbb{N}$ है। तो $f_{21}-f_{20}$ बराबर है _____

Ans. 41

Sol. $f_n(x)=\int_0^{\frac{\pi}{2}}\left(1+\sin x+\sin^2 x+\sin^3 x+...+\sin^{n-1}(x)\right)\left(1+3\sin x+5\sin^2 x+..+(2n-1)\right)\sin^{n-1} x.\cos x dx$

Multiply & divide by $\sqrt{\sin x}$

$$\int_0^{\frac{\pi}{2}}\left((\sin x)^{\frac{1}{2}}+(\sin x)^{\frac{3}{2}}+(\sin x)^{\frac{5}{2}}+(\sin x)^{\frac{7}{2}}+...(\sin x)^{\frac{2n-1}{2}}\right)$$

$$\left(1+3\sin x+5\sin^2 x+...+(2n-1)\sin^{n-1}(x)\right)\frac{\cos x}{\sqrt{\sin x}}dx$$

Put $(\sin x)^{1/2}+(\sin x)^{3/2}+(\sin x)^{5/2}+....(\sin x)^{n-1/2}=t$

$$\frac{1}{2}\frac{\left(1+3\sin x+5\sin^2 x+...(2n-1)\sin^{n-1} x\right)}{\sqrt{\sin x}}\cos x dx=dt$$

$$f_n=2\int_0^n t dt$$

$f_n=n^2$

$f_{21}-f_{20}=(21)^2-(20)^2$

$=441-400$

$=41$